

CHAPTER - 1. RELATIONS AND FUNCTIONS

PUBLIC EXAM IMPORTANT QUESTIONS-2026

2 MARK IMPORTANT QUESTIONS

1. If $A = \{1,3,5\}$ and $B = \{2,3\}$ then (i) find $A \times B$ and $B \times A$. **JUL-21**

2. If $A \times B = \{(3,2), (3,4), (5,2), (5,4)\}$ then find A and B . **APR-23, PTA-4**

3. Find $A \times B, A \times A$ and $B \times A$

(i) $A = \{2, -2, 3\}$ and $B = \{1, -4\}$ **JUL-25** (ii) $A = \{m, n\}; B = \phi$ **PTA-1**

4. Let $A = \{1,2,3\}$ and $B = \{x \mid x \text{ is a prime number less than } 10\}$. Find $A \times B$ and $B \times A$.

APR-22,25

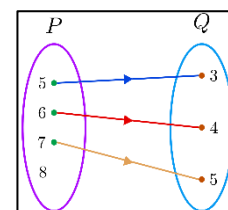
5. If $B \times A = \{(-2,3), (-2,4), (0,3), (0,4), (3,3), (3,4)\}$ find A and B .

JUL-24, APR-23

6. The arrow diagram shows (Fig.1.10) a relationship between the sets P and

Q . Write the relation in (i) Set builder form (ii) Roster form

(iii) What is the domain and range of R . **APR-22, JUL-25**



7. Let $A = \{1,2,3,4, \dots, 45\}$ and R be the relation defined as "square is of a number" on A .

Write R as a subset of $A \times A$. Also, find the domain and range of R .

JUL-21

8. A Relation R is given by the set $\{(x, y) / y = x + 3, x \in \{0,1,2,3,4,5\}\}$. Determine its domain and range. **JUL-23, PTA-5**

9. Represent each of the given relations by (a) an arrow diagram, (b) a graph and (c) a set in roster form, wherever possible. **JUL-22**

(i) $\{(x, y) \mid x = 2y, x \in \{2,3,4,5\}, y \in \{1,2,3,4\}\}$

(ii) $\{(x, y) \mid y = x + 3, x, y \text{ are natural numbers } < 10\}$

10. Let $X = \{1,2,3,4\}$ and $Y = \{2,4,6,8,10\}$ and $R = \{(1,2), (2,4), (3,6), (4,8)\}$. Show that R is a function and find its domain, co-domain and range?

11. A relation $f: X \rightarrow Y$ is defined by $f(x) = x^2 - 2$ where, $X = \{-2, -1, 0, 3\}$ and $Y = R$.

(i) List the elements of f (ii) Is f a function? **PTA-1**

12. Given $f(x) = 2x - x^2$, find (i) $f(1)$ (ii) $f(x + 1)$ (iii) $f(x) + f(1)$ **JUL-24**

13. Given the function $f: x \rightarrow x^2 - 5x + 6$, evaluate

(i) $f(-1)$ (ii) $f(2a)$ (iii) $f(2)$ (iv) $f(x - 1)$

14. Let $f(x) = 2x + 5$. If $x \neq 0$ then find $\frac{f(x+2)-f(2)}{x}$.

15. A function f is defined by $f(x) = 3 - 2x$. Find x such that $f(x^2) = (f(x))^2$.

16. A plane is flying at a speed of 500 km per hour. Express the distance ' d ' travelled by the plane as function of time t in hours.

10 th MATHS PUBLIC EXAM PREPARATION-2026

17. If $A = \{-2, -1, 0, 1, 2\}$ and $f: A \rightarrow B$ is an onto function defined by $f(x) = x^2 + x + 1$ then find B .
18. Let f be a function from $\mathbb{R}$ to $\mathbb{R}$ defined by $f(x) = 3x - 5$. Find the values of a and b given that $(a, 4)$ and $(1, b)$ belong to f . **PTA-6**
19. Show that the function $f: \mathbb{N} \rightarrow \mathbb{N}$ defined by $f(m) = m^2 + m + 3$ is one-one function.
20. Let $A = \{1, 2, 3, 4\}$ and $B = \mathbb{N}$. Let $f: A \rightarrow B$ be defined by $f(x) = x^3$ then,
(i) find the range of f (ii) identify the type of function
21. Find $f \circ g$ and $g \circ f$ when $f(x) = 2x + 1$ **APR-25**
22. If $f(x) = 3x - 2$, $g(x) = 2x + k$ and if $f \circ g = g \circ f$, then find the value of k .
23. Find k if $f \circ f(k) = 5$ where $f(k) = 2k - 1$. **APR-23, PTA-4**
24. Using the functions f and g given below, find $f \circ g$ and $g \circ f$.
Check whether $f \circ g = g \circ f$.
(i) $f(x) = x - 6$, $g(x) = x^2$ **JUL-23** (ii) $f(x) = \frac{2}{x}$, $g(x) = 2x^2 - 1$
(iii) $f(x) = \frac{x+6}{3}$, $g(x) = 3 - x$ (iv) $f(x) = 3 + x$, $g(x) = x - 4$ **MDL-19**
(v) $f(x) = 4x^2 - 1$, $g(x) = 1 + x$
25. Find the value of k , such that $f \circ g = g \circ f$
(i) $f(x) = 3x + 2$, $g(x) = 6x - k$ (ii) $f(x) = 2x - k$, $g(x) = 4x + 5$
26. If $f(x) = 2x - 1$, $g(x) = \frac{x+1}{2}$, show that $f \circ g = g \circ f = x$
27. If $f(x) = x^2 - 1$, $g(x) = x - 2$ find a , if $g \circ f(a) = 1$. **PTA-2**
28. Let $f(x) = x^2 - 1$. Find (i) $f \circ f$ (ii) $f \circ f \circ f$
29. If $f: \mathbb{R} \rightarrow \mathbb{R}$ and $g: \mathbb{R} \rightarrow \mathbb{R}$ are defined by $f(x) = x^5$ and $g(x) = x^4$ then check if f, g are one-one and $f \circ g$ is one-one? **PTA-6**
30. Given that $f(x) = \begin{cases} \sqrt{x-1} & x \geq 1 \\ 4 & x < 1 \end{cases}$. Find
(i) $f(0)$ (ii) $f(3)$ (iii) $f(a+1)$ in terms of a . (Given that $a \geq 0$)
31. Find the domain of the function $f(x) = \sqrt{1 + \sqrt{1 - \sqrt{1 - x^2}}}$
32. If $f(x) = \frac{x-1}{x+1}$, $x \neq -1$ show that $f(f(x)) = -\frac{1}{x}$, provided $x \neq 0$

5 MARK IMPORTANT QUESTIONS

1. Let $A = \{x \in \mathbb{N} \mid 1 < x < 4\}$, $B = \{x \in \mathbb{W} \mid 0 \leq x < 2\}$ and $C = \{x \in \mathbb{N} \mid x < 3\}$.

Then verify that (i) $A \times (B \cup C) = (A \times B) \cup (A \times C)$ **APR-24**

(ii) $A \times (B \cap C) = (A \times B) \cap (A \times C)$

2. Given $A = \{1,2,3\}$, $B = \{2,3,5\}$, $C = \{3,4\}$ and $D = \{1,3,5\}$, check if

$(A \cap C) \times (B \cap D) = (A \times B) \cap (C \times D)$ is true?

3. Let $A = \{x \in \mathbb{W} \mid x < 2\}$, $B = \{x \in \mathbb{N} \mid 1 < x \leq 4\}$ and $C = \{3,5\}$. Verify that

(i) $A \times (B \cup C) = (A \times B) \cup (A \times C)$ **JUL-23, PTA-2**

(ii) $A \times (B \cap C) = (A \times B) \cap (A \times C)$ **JUL-21, PTA-5**

(iii) $(A \cup B) \times C = (A \times C) \cup (B \times C)$

4. Let A = The set of all natural numbers less than 8, B = The set of all prime numbers less than 8, C = The set of even prime number. Verify that

(i) $(A \cap B) \times C = (A \times C) \cap (B \times C)$

(ii) $A \times (B - C) = (A \times B) - (A \times C)$. **APR-22, PTA-1** **JUL-20, 24, APR-25**

5. If $X = \{-5, 1, 3, 4\}$ and $Y = \{a, b, c\}$, then which of the following relations are functions from X to Y ?

(i) $R_1 = \{(-5, a), (1, a), (3, b)\}$ (ii) $R_2 = \{(-5, b), (1, b), (3, a), (4, c)\}$

(iii) $R_3 = \{(-5, a), (1, a), (3, b), (4, c), (1, b)\}$

6. A function f is defined by $f(x) = 2x - 3$

(i) find $\frac{f(0)+f(1)}{2}$. (ii) find x such that $f(x) = 0$.

(iii) find x such that $f(x) = x$.

(iv) find x such that $f(x) = f(1 - x)$

7. The data in the adjacent table depicts the length of a person forehead and their corresponding height. Based on this data, a student finds a relationship between the height (y) and the forehead length (x) as $y = ax + b$, where a, b are constants.

(i) Check if this relation is a function. (ii) Find a and b .

(iii) Find the height of a person whose forehead length is 40 cm

(iv) Find the length of forehead of a person if the height is 53.3 inches. **PTA-4**

Length 'x' of Forehead (in cm)	Height 'y' of (in inch)
35	56
45	65
50	69.5
55	74

8. Let $A = \{1,2,3,4\}$ and $B = \{2,5,8,11,14\}$ be two sets. Let $f: A \rightarrow B$ be a function given by

$f(x) = 3x - 1$. Represent this function **PTA-3, JUL-20, 25**

(i) by arrow diagram (ii) in a table form (iii) as a set of ordered pairs

(iv) in a graphical form

10 th MATHS PUBLIC EXAM PREPARATION-2026

9. Let f be a function $f: \mathbb{N} \rightarrow \mathbb{N}$ be defined by $f(x) = 3x + 2, x \in \mathbb{N}$ **PTA-3**

- (i) Find the images of 1,2,3 (ii) Find the pre-images of 29,53
(ii) Identify the type of function

10. If the function $f: \mathbb{R} \rightarrow \mathbb{R}$ is defined by $f(x) = \begin{cases} 2x + 7; & x < -2 \\ x^2 - 2; & -2 \leq x < 3, \\ 3x - 2; & x \geq 3 \end{cases}$

then find the values of

- (i) $f(4)$ (ii) $f(-2)$ (iii) $f(4) + 2f(1)$ (iv) $\frac{f(1)-3f(4)}{f(-3)}$

11. Let $f: A \rightarrow B$ be a function defined by $f(x) = \frac{x}{2} - 1$, where $A = \{2,4,6,10,12\}$,

$B = \{0,1,2,4,5,9\}$. Represent f by **APR-23, MDL-19**

- (i) set of ordered pairs (ii) a table (iii) an arrow diagram
(iv) a graph

12. Represent the function $f = \{(1,2), (2,2), (3,2), (4,3), (5,4)\}$ through

- (i) an arrow diagram (ii) a table form (iii) a graph

13. If the function f is defined by $f(x) = \begin{cases} x + 2; & x > 1 \\ 2; & -1 \leq x \leq 1 \\ x - 1; & -3 < x < -1 \end{cases}$ find the values of

- (i) $f(3)$ (ii) $f(0)$ (iii) $f(-1.5)$ (iv) $f(2) + f(-2)$

14. A function $f: [-5,9] \rightarrow \mathbb{R}$ is defined as follows: **JUL-24, PTA-4**

$$f(x) = \begin{cases} 6x + 1; & -5 \leq x < 2 \\ 5x^2 - 1; & 2 \leq x < 6 \\ 3x - 4; & 6 \leq x \leq 9 \end{cases}$$

Find (i) $f(-3) + f(2)$ (ii) $f(7) - f(1)$ (iii) $2f(4) + f(8)$ (iv) $\frac{2f(-2)-f(6)}{f(4)+f(-2)}$

15. The distance S an object travels under the influence of gravity in time t seconds is given by $S(t) = \frac{1}{2}gt^2 + at + b$ where, (g is the acceleration due to gravity), $a > 0, b$ are constants.

Verify wheather the function $S(t)$ is one-one or not. **PTA-3**

16. The function ' t ' which maps temperature in Celsius (C) into temperature in Fahrenheit (F) is defined by $t(C) = F$ where $F = \frac{9}{5}C + 32$. Find, **PTA-1**

- (i) $t(0)$ (ii) $t(28)$ (iii) $t(-10)$ (iv) the value of C when $t(C) = 212$
(v) the temperature when the Celsius value is equal to the Faren-heit value.

17. If $f(x) = 2x + 3, g(x) = 1 - 2x$ and $h(x) = 3x$. Prove that $f \circ (g \circ h) = (f \circ g) \circ h$

18. Find x if $gff(x) = fgg(x)$, given $f(x) = 3x + 1$ and $g(x) = x + 3$ **MDL-19**

19. Consider the functions $f(x), g(x), h(x)$ as given below. Show that $(f \circ g) \circ h = f \circ (g \circ h)$ in each case.

10 th MATHS PUBLIC EXAM PREPARATION-2026

(i) $f(x) = x - 1$, $g(x) = 3x + 1$ and $h(x) = x^2$ **JUL-25**

(ii) $f(x) = x^2$, $g(x) = 2x$ and $h(x) = x + 4$

(iii) $f(x) = x - 4$, $g(x) = x^2$ and $h(x) = 3x - 5$ **PTA-2**

20. If $f(x) = x^2$, $g(x) = 3x$ and $h(x) = x - 2$, Prove that $(f \circ g) \circ h = f \circ (g \circ h)$

21. The functions f and g are defined by $f(x) = 6x + 8$; $g(x) = \frac{x-2}{3}$

(i) Calculate the value of $gg\left(\frac{1}{2}\right)$

(ii) Write an expression for $gf(x)$ in its simplest form.

22. Write the domain of the following real functions **PTA-6**

(i) $f(x) = \frac{2x+1}{x-9}$ **PTA-6**

(ii) $p(x) = \frac{-5}{4x^2+1}$

(iii) $g(x) = \sqrt{x-2}$ **PTA-6**

(iv) $h(x) = x + 6$

CHAPTER - 2. NUMBERS AND SEQUENCES

PUBLIC EXAM IMPORTANT QUESTIONS-2026

2 MARK IMPORTANT QUESTIONS

1. If the Highest Common Factor of 210 and 55 is expressible in the form $55x - 325$, find x .
2. Find the greatest number that will divide 445 and 572 leaving remainders 4 and 5 respectively.
3. A man has 532 flower pots. He wants to arrange them in rows such that each row contains 21 flower pots. Find the number of completed rows and how many flower pots are left over. **APR-25**
4. Prove that the product of two consecutive positive integers is divisible by 2. **PTA-1**
5. Find the largest number which divides 1230 and 1926 leaving remainder 12 in each case. **JUL-22**
6. If d is the Highest Common Factor of 32 and 60, find x and y satisfying $d = 32x + 60y$.
7. Is $7 \times 5 \times 3 \times 2 + 3$ a composite number? Justify your Answer. **PTA-3**
8. 'a' and 'b' are two positive integers such that $a^b \times b^a = 800$. Find 'a' and 'b'. **APR-24, JUL-25**
9. If m, n are natural numbers, for what values of m , does $2^n \times 5^m$ ends in 5? **JUL-20**
10. Find the HCF of 252525 and 363636.
11. If $13824 = 2^a \times 3^b$ then find a and b . **APR-2**
12. Find the least number that is divisible by the first ten natural numbers. **JUL-22, 23**

10 th MATHS PUBLIC EXAM PREPARATION-2026

13. Find the least positive value of x such that (i) $67 + x \equiv 1(\text{mod}4)$ **PTA-2**
14. Solve $8x \equiv 1(\text{mod} 11)$
15. Compute x , such that $10^4 \equiv x(\text{mod}19)$ **MDL-19**
16. Find the number of integer solutions of $3x \equiv 1(\text{mod}15)$. **JUL-21, PTA-5**
17. Find the least positive value of x such that (i) $5x \equiv 4(\text{mod} 6)$ **PTA-2**
18. What is the time 100 hours after 7 a.m.?
19. What is the time 15 hours before 11 p.m.?
20. Today is Tuesday. My uncle will come after 45 days. In which day my uncle will be coming?
21. The general term of a sequence is defined as $a_n = \begin{cases} n(n+3) & ; n \in \mathbb{N} \text{ is odd} \\ n^2 + 1 & ; n \in \mathbb{N} \text{ is even} \end{cases}$ Find the eleventh and eighteenth terms.
22. Find a_8 and a_{15} whose n^{th} term is $a_n = \begin{cases} \frac{n^2-1}{n+3} & ; n \text{ is even, } n \in \mathbb{N} \\ \frac{n^2}{2n+1} & ; n \text{ is odd, } n \in \mathbb{N} \end{cases}$
23. Find the indicated terms of the sequences whose n^{th} terms are given by
(i) $a_n = \frac{5n}{n+2}$; a_6 and a_{13}
24. Check whether the following sequences are in A.P. or not? $3\sqrt{2}, 5\sqrt{2}, 7\sqrt{2}, 9\sqrt{2}, \dots$
25. Write an A.P. whose first term is 20 and common difference is 8.
26. Find the number of terms in the A.P. 3, 6, 9, 12, ..., 111. **JUL-21**
27. Find the 19^{th} term of an A.P. $-11, -15, -19, \dots$ **JUL-22, MDL-19**
28. Which term of an A.P. 16, 11, 6, 1, ... is -54 ? **JUL-24, APR-22**
29. Find the middle term(s) of an A.P. 9, 15, 21, 27, ..., 183. **PTA-1**
30. If $3 + k, 18 - k, 5k + 1$ are in A.P. then find k . **JUL-21, PTA-3,**
31. In a theatre, there are 20 seats in the front row and 30 rows were allotted. Each successive row contains two additional seats than its front row. How many seats are there in the last row? **PTA-4**
32. In a winter season let us take the temperature of Ooty from Monday to Friday to be in A.P. The sum of temperatures from Monday to Wednesday is 0°C and the sum of the temperatures from Wednesday to Friday is 18°C . Find the temperature on each of the five days.
33. Find the sum of $0.40 + 0.43 + 0.46 + \dots + 1$.
34. Find the sum of the $6 + 13 + 20 + \dots + 97$.
35. Find the sum of all odd positive integers less than 450.

10 th MATHS PUBLIC EXAM PREPARATION-2026

36. Find the geometric progression whose first term and common ratios are given by $a = -7$, $r = 6$ **PTA-5**
37. Find the 8th term of the G.P. 9, 3, 1, ... **JUL-23, 24**
38. In a G.P. 729, 243, 81, ... find t_7 . **JUL-25**
39. Find x so that $x + 6$, $x + 12$ and $x + 15$ are consecutive terms of a Geometric Progression.
40. Find the number of terms in the following G.P. (i) 4, 8, 16, ..., 8192 ?
41. If a, b, c are in A.P. then show that $3^a, 3^b, 3^c$ are in G.P
42. Find the sum of 8 terms of the G.P. 1, -3, 9, -27 ...
43. Find the sum $3 + 1 + \frac{1}{3} + \dots \infty$
44. Find the rational form of the number 0.6666 ...
45. Find the sum to infinity of (i) $9 + 3 + 1 + \dots$ (ii) $21 + 14 + \frac{28}{3} + \dots$
46. Find the sum of first six terms of the G.P. 5, 15, 45, ...
47. Find the rational form of the number $0.\overline{123}$.
48. Find the sum of (i) $1 + 3 + 5 + \dots$ to 40 terms (ii) $2 + 4 + 6 + \dots + 80$
(iii) $1 + 3 + 5 + \dots + 55$ **PTA-6**
49. If $1 + 2 + 3 + \dots + n = 666$ then find n . **PTA-2**
50. If $1 + 2 + 3 + \dots + k = 325$, then find $1^3 + 2^3 + 3^3 + \dots + k^3$.
51. If $1^3 + 2^3 + 3^3 + \dots + k^3 = 44100$ then find $1 + 2 + 3 + \dots + k$.
52. Prove that $n^2 - n$ divisible by 2 for every positive integer n .
53. Find the 12th term from the last term of the A. P $-2, -4, -6, \dots - 100$.
54. Show that 107 is of the form $4q + 3$ for any integer q .
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5 MARK IMPORTANT QUESTIONS

1. Find the HCF of 396, 504, 636. **JUL-21, 25**
2. Use Euclid's Division Algorithm to find the Highest Common Factor (HCF) of
(i) 340 and 412 (ii) 867 and 255
(iii) 10224 and 9648 (iv) 84, 90 and 120
3. If $p_1^{x_1} \times p_2^{x_2} \times p_3^{x_3} \times p_4^{x_4} = 113400$ where p_1, p_2, p_3, p_4 are primes in ascending order and x_1, x_2, x_3, x_4 are integers, find the value of p_1, p_2, p_3, p_4 and x_1, x_2, x_3, x_4 . **APR-25**
4. Find the greatest number consisting of 6 digits which is exactly divisible by 24, 15, 36 ?
5. Kala and Vani are friends. Kala says, "Today is my birthday" and she asks Vani, "When will you celebrate your birthday?" Vani replies, "Today is Monday and I celebrated my birthday 75 days ago". Find the day when Vani celebrated her birthday.

10 th MATHS PUBLIC EXAM PREPARATION-2026

6. Prove that $2^n + 6 \times 9^n$ is always divisible by 7 for any positive integer n .
7. Find the remainder when 2^{81} is divided by 17 .
8. The duration of flight travel from Chennai to London through British Airlines is approximately 11 hours. The airplane begins its journey on Sunday at 23:30 hours. If the time at Chennai is four and half hours ahead to that of London's time, then find the time at London, when will the flight lands at London Airport.
9. Find the first five terms of the following sequence.

$$a_1 = 1, a_2 = 1, a_n = \frac{a_{n-1}}{a_{n-2} + 3}; n \geq 3, n \in \mathbb{N}$$

10. If $l^{\text{th}}, m^{\text{th}}$ and n^{th} terms of an A.P. are x, y, z respectively, then show that **APR-22**
(i) $x(m - n) + y(n - l) + z(l - m) = 0$ (ii) $(x - y)n + (y - z)l + (z - x)m = 0$
11. In an A.P., sum of four consecutive terms is 28 and the sum of their squares is 276 . Find the four numbers.
12. A mother divides ₹207 into three parts such that the amount are in A.P. and gives it to her three children. The product of the two least amounts that the children had ₹ 4623 . Find the amount received by each child.
13. If nine times ninth term is equal to the fifteen times fifteenth term, show that six times twenty fourth term is zero. **JUL-22**
14. Find x, y and z , given that the numbers $x, 10, y, 24, z$ are in A.P.
15. The sum of three consecutive terms that are in A.P. is 27 and their product is 288 . Find the three terms.
16. The ratio of 6^{th} and 8^{th} term of an A.P. is 7: 9. Find the ratio of 9^{th} term to 13^{th} term. **APR-22, MDL-19**
17. The 13^{th} term of an A.P. is 3 and the sum of first 13 terms is 234 . Find the common difference and the sum of first 21 terms.
18. Find the sum of all natural numbers between 300 and 600 which are divisible by 7 .
19. The houses of a street are numbered from 1 to 49 . Senthil's house is numbered such that the sum of numbers of the houses prior to Senthil's house is equal to the sum of numbers of the houses following Senthil's house. Find Senthil's house number? **APR-23, PTA-2**
20. The sum of first $n, 2n$ and $3n$ terms of an A.P. are S_1, S_2 and S_3 respectively.
Prove that $S_3 = 3(S_2 - S_1)$. **MDL-19**
21. The 104^{th} term and 4^{th} term of an A.P. are 125 and 0 . Find the sum of first 35 terms.
22. A man repays a loan of ₹ 65,000 by paying ₹ 400 in the first month and then increasing the payment by ₹ 300 every month. How long will it take for him to clear the loan?

10 th MATHS PUBLIC EXAM PREPARATION-2026

23. If $S_1, S_2, S_3, \dots, S_m$ are the sums of n terms of m A.P.'s whose first terms are $1, 2, 3, \dots, m$ and whose common differences are $1, 3, 5, \dots, (2m - 1)$ respectively, then
Show that $S_1 + S_2 + S_3 + \dots + S_m = \frac{1}{2}mn(mn + 1)$
24. Find the sum $\left[\frac{a-b}{a+b} + \frac{3a-2b}{a+b} + \frac{5a-3b}{a+b} + \dots\right]$ to 12 terms].
25. In a Geometric progression, the 4th term is $\frac{8}{9}$ and the 7th term is $\frac{64}{243}$. Find the Geometric Progression.
26. The product of three consecutive terms of a Geometric Progression is 343 and their sum is $\frac{91}{3}$. Find the three terms.
27. In a G.P. the 9th term is 32805 and 6th term is 1215 . Find the 12th term.
28. In a G.P. the product of three consecutive terms is 27 and the sum of the product of two terms taken at a time is $\frac{57}{2}$. Find the three terms.
29. A man joined a company as Assistant Manager. The company gave him a starting salary of ₹ 60,000 and agreed to increase his salary 5% annually. What will be his salary after 5 years?
- PTA-6**
30. If a, b, c are three consecutive terms of an A.P. and x, y, z are three consecutive terms of a G.P. then prove that $x^{b-c} \times y^{c-a} \times z^{a-b} = 1$.
31. Find the sum to n terms of the series $5 + 55 + 555 + \dots$ **APR-23, PTA-4**
32. If the first term of an infinite G.P. is 8 and its sum to infinity is $\frac{32}{3}$ then find the common ratio.
33. Find the sum to n terms of the series
(i) $0.4 + 0.44 + 0.444 + \dots$ to n terms (ii) $3 + 33 + 333 + \dots$ to n terms **JUL-23, 24**
34. Find the sum of the Geometric series $3 + 6 + 12 + \dots + 1536$. **PTA-3**
35. If $S_n = (x + y) + (x^2 + xy + y^2) + (x^3 + x^2y + xy^2 + y^3) + \dots$ n terms then prove that
PTA-1
$$(x - y)S_n = \left[\frac{x^2(x^n - 1)}{x - 1} - \frac{y^2(y^n - 1)}{y - 1} \right]$$
36. Find the sum of (i) $5^2 + 10^2 + 15^2 + \dots + 105^2$ (ii) $15^2 + 16^2 + 17^2 + \dots + 28^2$
37. Find the sum of (i) $9^3 + 10^3 + \dots + 21^3$ **APR-24**
38. Find the sum of (i) $1 + 4 + 9 + 16 + \dots + 225$ (ii) $6^2 + 7^2 + 8^2 + \dots + 21^2$ **APR-25**
(iii) $10^3 + 11^3 + 12^3 + \dots + 20^3$ **PTA-5**
39. The sum of the cubes of the first n natural numbers is 2025 , then find the value of n .
40. Rekha has 15 square colour papers of sizes 10 cm, 11 cm, 12 cm, ..., 24 cm. How much area can be decorated with these colour papers? **JUN-23, 24, 25, PTA-1**
41. Find the sum of the series $(2^3 - 1^3) + (4^3 - 3^3) + (6^3 - 5^3) + \dots$ to
(i) n terms (ii) 8 terms

10 th MATHS PUBLIC EXAM PREPARATION-2026

42. A milk man has 175 litres of cow's milk and 105 litres of buffalow's milk. He wishes to sell the milk by filling the two types of milk in cans of equal capacity. Calculate the following
(i) Capacity of a can (ii) Number of cans of cow's milk
(iii) Number of cans of buffalow's milk.
43. A man saved ₹ 16500 in ten years. In each year after the first he saved ₹ 100 more than he did in the preceding year. How much did he save in the first year? **PTA-4**
44. Find the G.P. in which the 2nd term is $\sqrt{6}$ and the 6th term is $9\sqrt{6}$.

CHAPTER - 3. ALGEBRA

PUBLIC EXAM IMPORTANT QUESTIONS-2026

2 MARK IMPORTANT QUESTIONS

1. The father's age is six times his son's age. Six years hence the age of father will be four times his son's age. Find the present ages (in years) of the son and father.
2. Solve $2x - 3y = 6, x + y = 1$
3. Find the LCM of the (i) $8x^4y^2, 48x^2y^4$ (ii) $x^3 - 27, (x - 3)^2, x^2 - 9$
4. Find the LCM of the (i) $9a^3b^2, 12a^2b^2c$ **JUL-24** (ii) $p^2 - 3p + 2, p^2 - 4$
(iii) $(2x^2 - 3xy)^2, (4x - 6y)^3, 8x^3 - 27y^3$
5. Find the LCM of polynomials $a^2 + 4a - 12, a^2 - 5a + 6$ whose GCD is $a - 2$ **PTA-6**
6. Find the excluded values of the following expressions
(i) $\frac{x+10}{8x}$ (ii) $\frac{7p+2}{8p^2+13p+5}$ (iii) $\frac{x}{x^2+1}$
7. Find the excluded values, if any of the following expressions.
(i) $\frac{t}{t^2-5t+6}$ **APR-22, JUL-25** (ii) $\frac{x^2+6x+8}{x^2+x-2}$ (iii) $\frac{x^3-27}{x^3+x^2-6x}$
8. Simplify (i) $\frac{4x^2y}{2z^2} \times \frac{6xz^3}{20y^4}$ **APR-24, MDL-19** (ii) $\frac{p^2-10p+21}{p-7} \times \frac{p^2+p-12}{(p-3)^2}$
9. Simplify $\frac{x^3-y^3}{3x^2+9xy+6y^2} \times \frac{x^2+2xy+y^2}{x^2-y^2}$
10. Simplify $\frac{x+2}{4y} \div \frac{x^2-x-6}{12y^2}$ **APR-23**
11. If a polynomial $p(x) = x^2 - 5x - 14$ is divided by another polynomial $q(x)$ we get $\frac{x-7}{x+2}$, find $q(x)$. **PTA-2**
12. Simplify (i) $\frac{x(x+1)}{x-2} + \frac{x(1-x)}{x-2}$ (ii) $\frac{x+2}{x+3} + \frac{x-1}{x-2}$ (iii) $\frac{x^3}{x-y} + \frac{y^3}{y-x}$
13. Simplify (i) $\frac{(2x+1)(x-2)}{x-4} - \frac{(2x^2-5x+2)}{x-4}$ (ii) $\frac{4x}{x^2-1} - \frac{x+1}{x-1}$
14. Pari needs 4 hours to complete a work. His friend Yuvan needs 6 hours to complete the same work. How long will it take to complete if they work together? **MDL-19**
15. Find the square root of the following expressions
(i) $256(x-a)^8(x-b)^4(x-c)^{16}(x-d)^{20}$ (ii) $\frac{144a^8b^{12}c^{16}}{81f^{12}g^4h^{14}}$ **PTA-5**

10 th MATHS PUBLIC EXAM PREPARATION-2026

16. Find the square root of the following rational expressions.
 (i) $\frac{400x^4y^{12}z^{16}}{100x^8y^4z^4}$ **JUL-22** (ii) $\frac{121(a+b)^8(x+y)^8(b-c)^8}{81(b-c)^4(a-b)^{12}(b-c)^4}$
17. Find the sum and product of the roots (i) $x^2 + 8x - 65 = 0$ **APR-2** (ii) $kx^2 - k^2x - 2k^3 = 0$
18. Determine the quadratic equations, whose sum and product of roots are
 (i) $-9, 20$ **JUL-21** (ii) $\frac{-3}{2}, -1$ **PTA-4**
19. Find the sum and product of the roots (i) $x^2 + 3x - 28 = 0$
 (ii) $3 + \frac{1}{a} = \frac{10}{a^2}$ (iii) $3y^2 - y - 4 = 0$
20. Solve $2x^2 - 2\sqrt{6}x + 3 = 0$ **PTA-6**
21. Solve $2m^2 + 19m + 30 = 0$
22. Solve the following quadratic equations by factorization method
 (i) $3(p^2 - 6) = p(p + 5)$ (ii) $\sqrt{a(a-7)} = 3\sqrt{2}$ (iii) $\sqrt{2}x^2 + 7x + 5\sqrt{2} = 0$
23. Solve $x^2 + 2x - 2 = 0$ by formula method **JUL-22**
24. Solve $2x^2 - 3x - 3 = 0$ by formula method.
25. Solve the following quadratic equations by formula method
 (i) $\sqrt{2}f^2 - 6f + 3\sqrt{2} = 0$ (ii) $3y^2 - 20y - 23 = 0$
26. The product of Kumaran's age (in years) two years ago and his age four years from now is one more than twice his present age. What is his present age? **PTA-1**
27. If the difference between a number and its reciprocal is $\frac{24}{5}$, find the number. **PTA-6**
28. Determine the nature of roots for the following quadratic equations
 (i) $x^2 - x - 20 = 0$ (ii) $9x^2 - 24x + 16 = 0$ (iii) $2x^2 - 2x + 9 = 0$
29. Determine the nature of the roots for the following quadratic equations
 (i) $15x^2 + 11x + 2 = 0$ **JUL-21, 23, 24** (ii) $x^2 - x - 1 = 0$
 (iii) $\sqrt{2}t^2 - 3t + 3\sqrt{2} = 0$ (iv) $9a^2b^2x^2 - 24abcdx + 16c^2d^2 = 0, a \neq 0, b \neq 0$
30. If the difference between the roots of the equation $x^2 - 13x + k = 0$ is 17 find k .
31. Write each of the following expression in terms of $\alpha + \beta$ and $\alpha\beta$
 (i) $\frac{\alpha}{3\beta} + \frac{\beta}{3\alpha}$ (ii) $\frac{1}{\alpha^2\beta} + \frac{1}{\beta^2\alpha}$
 (iii) $(3\alpha - 1)(3\beta - 1)$ (iv) $\frac{\alpha+3}{\beta} + \frac{\beta+3}{\alpha}$
32. If a matrix has 16 elements, what are the possible orders it can have?
33. Construct a 3×3 matrix whose elements are $a_{ij} = i^2j^2$
34. In the matrix $A = \begin{pmatrix} 8 & 9 & 4 & 3 \\ -1 & \sqrt{7} & \frac{\sqrt{3}}{2} & 5 \\ 1 & 4 & 3 & 0 \\ 6 & 8 & -11 & 1 \end{pmatrix}$, write (i) The number of elements
 (ii) The order of the matrix (iii) Write the elements $a_{22}, a_{23}, a_{24}, a_{34}, a_{43}, a_{44}$
35. Construct a 3×3 matrix whose elements are given by
 (i) $a_{ij} = |i - 2j|$
36. If $A = \begin{pmatrix} \sqrt{7} & -3 \\ -\sqrt{5} & 2 \\ \sqrt{3} & -5 \end{pmatrix}$ then find the transpose of $-A$. **JUL- 20, PTA-2**

37. If $A = \begin{pmatrix} 5 & 2 & 2 \\ -\sqrt{17} & 0.7 & \frac{5}{2} \\ 8 & 3 & 1 \end{pmatrix}$ then verify $(A^T)^T = A$ **JUL- 2**
38. Find the values of x, y and z from the following equations
 (i) $\begin{pmatrix} 12 & 3 \\ x & 5 \end{pmatrix} = \begin{pmatrix} y & z \\ 3 & 5 \end{pmatrix}$ **MDL-19** (ii) $\begin{pmatrix} x+y & 2 \\ 5+z & xy \end{pmatrix} = \begin{pmatrix} 6 & 2 \\ 5 & 8 \end{pmatrix}$
39. If $A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}, B = \begin{pmatrix} 1 & 7 & 0 \\ 1 & 3 & 1 \\ 2 & 4 & 0 \end{pmatrix}$, find $A + B$.
40. If $A = \begin{pmatrix} 1 & 3 & -2 \\ 5 & -4 & 6 \\ -3 & 2 & 9 \end{pmatrix}, B = \begin{pmatrix} 1 & 8 \\ 3 & 4 \\ 9 & 6 \end{pmatrix}$, find $A + B$.
41. If $A = \begin{pmatrix} 7 & 8 & 6 \\ 1 & 3 & 9 \\ -4 & 3 & -1 \end{pmatrix}, B = \begin{pmatrix} 4 & 11 & -3 \\ -1 & 2 & 4 \\ 7 & 5 & 0 \end{pmatrix}$ then Find $2A + B$. **PTA-3**
42. If $A = \begin{pmatrix} 1 & 9 \\ 3 & 4 \\ 8 & -3 \end{pmatrix}, B = \begin{pmatrix} 5 & 7 \\ 3 & 3 \\ 1 & 0 \end{pmatrix}$ then verify that
 (i) $A + B = B + A$ (ii) $A + (-A) = (-A) + A = O$.
43. If $A = \begin{pmatrix} 0 & 4 & 9 \\ 8 & 3 & 7 \end{pmatrix}, B = \begin{pmatrix} 7 & 3 & 8 \\ 1 & 4 & 9 \end{pmatrix}$ find the value of (i) $B - 5A$ (ii) $3A - 9B$ **PTA-5**
44. If $A = \begin{pmatrix} 1 & 2 & 0 \\ 3 & 1 & 5 \end{pmatrix}, B = \begin{pmatrix} 8 & 3 & 1 \\ 2 & 4 & 1 \\ 5 & 3 & 1 \end{pmatrix}$, find AB .
45. If $A = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}, B = \begin{pmatrix} 2 & 0 \\ 1 & 3 \end{pmatrix}$ find AB and BA . Verify $AB = BA$? **APR- 25**
46. If $A = \begin{pmatrix} 2 & -2\sqrt{2} \\ \sqrt{2} & 2 \end{pmatrix}$ and $B = \begin{pmatrix} 2 & 2\sqrt{2} \\ -\sqrt{2} & 2 \end{pmatrix}$. Show that A and B satisfy commutative property with respect to matrix multiplication
47. A has ' a ' rows and ' $a + 3$ ' columns. B has ' b ' rows and ' $17 - b$ ' columns, and if both products AB and BA exist, find a, b ?
48. If A is of order $p \times q$ and B is of order $q \times r$ what is the order of AB and BA ? **PTA-1**
49. If $A = \begin{pmatrix} 2 & 5 \\ 4 & 3 \end{pmatrix}, B = \begin{pmatrix} 1 & -3 \\ 2 & 5 \end{pmatrix}$ find AB, BA and verify $AB = BA$?
50. If $A = \begin{pmatrix} \cos \theta & 0 \\ 0 & \cos \theta \end{pmatrix}, B = \begin{pmatrix} \sin \theta & 0 \\ 0 & \sin \theta \end{pmatrix}$ then show that $A^2 + B^2 = I$. **PTA-2**
51. Verify that $A^2 = I$ when $A = \begin{pmatrix} 5 & -4 \\ 6 & -5 \end{pmatrix}$.
52. If -4 is a root of the equation $x^2 + px - 4 = 0$ and if the equation $x^2 + px + q = 0$ has equal roots, find the values of p and q .
53. If $\cos \theta \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix} + \sin \theta \begin{pmatrix} x & -\cos \theta \\ \cos \theta & x \end{pmatrix} = I_2$, find x .

5 MARK IMPORTANT QUESTIONS

1. Solve the following system of linear equations in three variables **JUL-23**
 $3x - 2y + z = 2, 2x + 3y - z = 5, x + y + z = 6.$
2. Solve $x + 2y - z = 5; x - y + z = -2; -5x - 4y + z = -11$
3. Solve $3x + y - 3z = 1; -2x - y + 2z = 1; -x - y + z = 2.$
4. Solve the following system of linear equations in three variables
 (i) $x + y + z = 5; 2x - y + z = 9; x - 2y + 3z = 16$ **JUL-21, PTA-5**
 (ii) $x + 20 = \frac{3y}{2} + 10 = 2z + 5 = 110 - (y + z)$ **APR-23**
5. Discuss the nature of solutions of the following system of equations

$$\frac{y+z}{4} = \frac{z+x}{3} = \frac{x+y}{2}; x + y + z = 27$$
6. Vani, her father and her grand father have an average age of 53 . One-half of her grand father's age plus one-third of her father's age plus one fourth of Vani's age is 65. Four years ago if Vani's grandfather was four times as old as Vani then how old are they all now? **PTA-2**
7. Find the GCD of the polynomials $x^3 + x^2 - x + 2$ and $2x^3 - 5x^2 + 5x - 3.$
8. Find the GCD of $6x^3 - 30x^2 + 60x - 48$ and $3x^3 - 12x^2 + 21x - 18.$
9. Find the GCD of the given polynomials
 (i) $x^4 + 3x^3 - x - 3, x^3 + x^2 - 5x + 3$ **JUL-20**
 (ii) $3x^4 + 6x^3 - 12x^2 - 24x, 4x^4 + 14x^3 + 8x^2 - 8x$
 (iii) $3x^3 + 3x^2 + 3x + 3, 6x^3 + 12x^2 + 6x + 12$
10. Find $\frac{16x^2-2x-3}{3x^2-2x-1} \div \frac{8x^2+11x+3}{3x^2-11x-4}$
11. Simplify: $\frac{b^2+3b-28}{b^2+4b+4} \div \frac{b^2-49}{b^2-5b-14}$ **JUL-22**
12. If $x = \frac{a^2+3a-4}{3a^2-3}$ and $y = \frac{a^2+2a-8}{2a^2-2a-4}$ find the value of $x^2y^{-2}.$ **PTA-3**
13. Simplify $\frac{1}{x^2-5x+6} + \frac{1}{x^2-3x+2} - \frac{1}{x^2-8x+15}$
14. If $A = \frac{2x+1}{2x-1}, B = \frac{2x-1}{2x+1}$ find $\frac{1}{A-B} - \frac{2B}{A^2-B^2}$
15. If $A = \frac{x}{x+1}, B = \frac{1}{x+1}$, prove that $\frac{(A+B)^2+(A-B)^2}{A \div B} = \frac{2(x^2+1)}{x(x+1)^2}$
16. Iniya bought 50 kg of fruits consisting of apples and bananas. She paid twice as much per kg for the apple as she did for the banana. If Iniya bought ₹ 1800 worth of apples and ₹ 600 worth bananas, then how many kgs of each fruit did she buy?
17. Find the square root of $64x^4 - 16x^3 + 17x^2 - 2x + 1$ **APR-24, JUL-21,24**
18. If $9x^4 + 12x^3 + 28x^2 + ax + b$ is a perfect square, find the values of a and $b.$ **PTA-5**
19. Find the square root of the following polynomials by division method
 (i) $x^4 - 12x^3 + 42x^2 - 36x + 9$ **JUL-22** (ii) $37x^2 - 28x^3 + 4x^4 + 42x + 9$
 (iii) $16x^4 + 8x^2 + 1$ (iv) $121x^4 - 198x^3 - 183x^2 + 216x + 144$ **JUL-23**
20. Find the values of a and b if the following polynomials are perfect squares
 (i) $4x^4 - 12x^3 + 37x^2 + bx + a$ **PTA-4** (ii) $ax^4 + bx^3 + 361x^2 + 220x + 100$
21. Find the values of m and n if the following polynomials are perfect squares
 (i) $36x^4 - 60x^3 + 61x^2 - mx + n$ **APR-22, JUL-25** (ii) $x^4 - 8x^3 + mx^2 + nx + 16$
22. Solve $pqx^2 - (p+q)^2x + (p+q)^2 = 0$ **APR-22**

10 th MATHS PUBLIC EXAM PREPARATION-2026

23. A bus covers a distance of 90 km at a uniform speed. Had the speed been 15 km/ hour more it would have taken 30 minutes less for the journey. Find the original speed of the bus.
24. A girl is twice as old as her sister. Five years hence, the product of their ages (in years) will be 375 . Find their present ages. **PTA-4**
25. From a group of $2x^2$ black bees, square root of half of the group went to a tree. Again eight-ninth of the bees went to the same tree. The remaining two got caught up in a fragrant lotus. How many bees were there in total?
26. The hypotenuse of a right angled triangle is 25 cm and its perimeter 56 cm . Find the length of the smallest side.
27. If the roots of $(a - b)x^2 + (b - c)x + (c - a) = 0$ are real and equal, then prove that b, a, c are in arithmetic progression.
28. If a, b are real then show that the roots of the equation $(a - b)x^2 - 6(a + b)x - 9(a - b) = 0$ are real and unequal.
29. If the roots of the equation $(c^2 - ab)x^2 - 2(a^2 - bc)x + b^2 - ac = 0$ are real and equal prove that either $a = 0$ (or) $a^3 + b^3 + c^3 = 3abc$ **PTA-6**
30. If α, β are the roots of the equation $3x^2 + 7x - 2 = 0$, find the values of
 (i) $\frac{\alpha}{\beta} + \frac{\beta}{\alpha}$ (ii) $\frac{\alpha^2}{\beta} + \frac{\beta^2}{\alpha}$
31. If α, β are the roots of the equation $2x^2 - x - 1 = 0$, then form the equation whose roots are
 (i) $\frac{1}{\alpha}, \frac{1}{\beta}$ (ii) $\alpha^2\beta, \beta^2\alpha$ **MDL-19** (iii) $2\alpha + \beta, 2\beta + \alpha$
32. The roots of the equation $2x^2 - 7x + 5 = 0$ are α and β . Without solving for the roots, find
 (i) $\frac{1}{\alpha} + \frac{1}{\beta}$ (ii) $\frac{\alpha}{\beta} + \frac{\beta}{\alpha}$ (iii) $\frac{\alpha+2}{\beta+2} + \frac{\beta+2}{\alpha+2}$
33. If α and β are the roots of $x^2 + 7x + 10 = 0$ find the values of
 (i) $(\alpha - \beta)$ (ii) $\alpha^2 + \beta^2$ (iii) $\alpha^3 - \beta^3$
 (iv) $\alpha^4 + \beta^4$ (v) $\frac{\alpha}{\beta} + \frac{\beta}{\alpha}$ (vi) $\frac{\alpha^2}{\beta} + \frac{\beta^2}{\alpha}$
34. If α, β are the roots of $7x^2 + ax + 2 = 0$ and if $\beta - \alpha = \frac{-13}{7}$. Find the values of a . **PTA-6, APR-22**
35. If one root of the equation $2y^2 - ay + 64 = 0$ is twice the other then find the values of a . **PTA-6**
36. If one root of the equation $3x^2 + kx + 81 = 0$ (having real roots) is the square of the other then find k .
37. Find the value of a, b, c, d from the equation $\begin{pmatrix} a-b & 2a+c \\ 2a-b & 3c+d \end{pmatrix} = \begin{pmatrix} 1 & 5 \\ 0 & 2 \end{pmatrix}$
38. If $A = \begin{pmatrix} 4 & 3 & 1 \\ 2 & 3 & -8 \\ 1 & 0 & -4 \end{pmatrix}, B = \begin{pmatrix} 2 & 3 & 4 \\ 1 & 9 & 2 \\ -7 & 1 & -1 \end{pmatrix}$ and $C = \begin{pmatrix} 8 & 3 & 4 \\ 1 & -2 & 3 \\ 2 & 4 & -1 \end{pmatrix}$ then verify that $A + (B + C) = (A + B) + C$.
39. Find X and Y if $X + Y = \begin{pmatrix} 7 & 0 \\ 3 & 5 \end{pmatrix}$ and $X - Y = \begin{pmatrix} 3 & 0 \\ 0 & 4 \end{pmatrix}$.
40. Find the non-zero values of x satisfying the matrix equation **PTA-4**

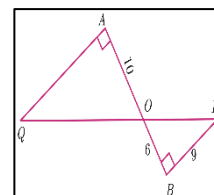
$$x \begin{pmatrix} 2x & 2 \\ 3 & x \end{pmatrix} + 2 \begin{pmatrix} 8 & 5x \\ 4 & 4x \end{pmatrix} = 2 \begin{pmatrix} x^2 + 8 & 24 \\ 10 & 6x \end{pmatrix}$$
41. Solve for x, y : $\begin{pmatrix} x^2 \\ y^2 \end{pmatrix} + 2 \begin{pmatrix} -2x \\ -y \end{pmatrix} = \begin{pmatrix} 5 \\ 8 \end{pmatrix}$

42. If $A = \begin{pmatrix} 1 & -1 & 2 \end{pmatrix}$, $B = \begin{pmatrix} 1 & -1 \\ 2 & 1 \\ 1 & 3 \end{pmatrix}$ and $C = \begin{pmatrix} 1 & 2 \\ 2 & -1 \end{pmatrix}$ show that $(AB)C = A(BC)$.
43. If $A = \begin{pmatrix} 1 & 1 \\ -1 & 3 \end{pmatrix}$, $B = \begin{pmatrix} 1 & 2 \\ -4 & 2 \end{pmatrix}$, $C = \begin{pmatrix} -7 & 6 \\ 3 & 2 \end{pmatrix}$ verify that $A(B + C) = AB + AC$. PTA- 1
44. If $A = \begin{pmatrix} 1 & 2 & 1 \\ 2 & -1 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} 2 & -1 \\ -1 & 4 \\ 0 & 2 \end{pmatrix}$ show that $(AB)^T = B^T A^T$ APR- 25, JUL-20
45. Given that $A = \begin{pmatrix} 1 & 3 \\ 5 & -1 \end{pmatrix}$, $B = \begin{pmatrix} 1 & -1 & 2 \\ 3 & 5 & 2 \end{pmatrix}$, $C = \begin{pmatrix} 1 & 3 & 2 \\ -4 & 1 & 3 \end{pmatrix}$ verify that $A(B + C) = AB + AC$.
46. If $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ and $I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ show that $A^2 - (a + d)A = (bc - ad)I_2$.
47. If $A = \begin{pmatrix} 5 & 2 & 9 \\ 1 & 2 & 8 \end{pmatrix}$, $B = \begin{pmatrix} 1 & 7 \\ 1 & 2 \\ 5 & -1 \end{pmatrix}$ verify that $(AB)^T = B^T A^T$ PTA-3, JUL-24, APR-23
48. If $A = \begin{pmatrix} 3 & 1 \\ -1 & 2 \end{pmatrix}$ show that $A^2 - 5A + 7I_2 = 0$ JUL-23, APR-24, JUL-2
49. Solve $\frac{1}{3}(x + y - 5) = y - z = 2x - 11 = 9 - (x + 2z)$.
50. Find the GCD of the following by division algorithm
 $2x^4 + 13x^3 + 27x^2 + 23x + 7$, $x^3 + 3x^2 + 3x + 1$, $x^2 + 2x + 1$.
51. Find the square root of $289x^4 - 612x^3 + 970x^2 - 684x + 361$.
52. A boat takes 1.6 hours longer to go 36 kms up a river than down the river. If the speed of the water current is 4 km per hr, what is the speed of the boat in still water?
53. If α and β are the roots of the polynomial $f(x) = x^2 - 2x + 3$, find the polynomial whose roots are (i) $\alpha + 2, \beta + 2$ (ii) $\frac{\alpha-1}{\alpha+1}, \frac{\beta-1}{\beta+1}$
54. Given $A = \begin{pmatrix} p & 0 \\ 0 & 2 \end{pmatrix}$, $B = \begin{pmatrix} 0 & -q \\ 1 & 0 \end{pmatrix}$, $C = \begin{pmatrix} 2 & -2 \\ 2 & 2 \end{pmatrix}$ and if $BA = C^2$, find p and q .
55. $A = \begin{pmatrix} 3 & 0 \\ 4 & 5 \end{pmatrix}$, $B = \begin{pmatrix} 6 & 3 \\ 8 & 5 \end{pmatrix}$, $C = \begin{pmatrix} 3 & 6 \\ 1 & 1 \end{pmatrix}$ find the matrix D , such that $CD - AB = 0$

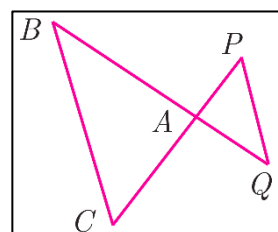
CHAPTER - 4. GEOMETRY

PUBLIC EXAM IMPORTANT QUESTIONS-2026

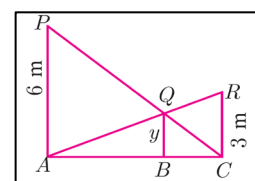
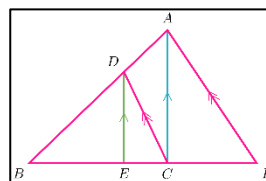
2 MARK IMPORTANT QUESTIONS



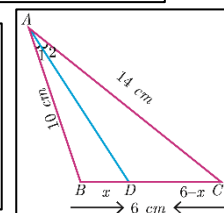
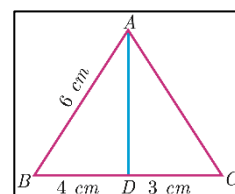
1. In Fig.4.21, QA and PB are perpendiculars to AB . If $AO = 10$ cm, $BO = 6$ cm and $PB = 9$ cm. Find AQ . **JUL-24**
2. The perimeters of two similar triangles ABC and PQR are respectively 36 cm and 24 cm. If $PQ = 10$ cm, find AB .
3. If $\triangle ABC$ is similar to $\triangle DEF$ such that $BC = 3$ cm, $EF = 4$ cm and area of $\triangle ABC = 54$ cm². Find the area of $\triangle DEF$. **PTA-2**



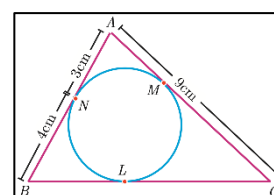
4. In the adjacent figure, $\triangle ACB \sim \triangle APQ$. If $BC = 8$ cm, $PQ = 4$ cm, $BA = 6.5$ cm and $AP = 2.8$ cm, find CA and AQ
5. Two vertical poles of heights 6 m and 3 m are erected above a horizontal ground AC . Find the value of y . **PTA-5**
6. In the Fig.4.38, $DE \parallel AC$ and $DC \parallel AP$. Prove that $\frac{BE}{EC} = \frac{BC}{CP}$. **PTA-4**



7. In the Fig.4.39, AD is the bisector of $\angle A$. If $BD = 4$ cm, $DC = 3$ cm and $AB = 6$ cm, find AC . **PTA-5, APR-22**
8. In the Fig. 4.40, AD is the bisector of $\angle BAC$, if $AB = 10$ cm, $AC = 14$ cm and $BC = 6$ cm. Find BD and DC . **PTA-3, APR-23**
9. In $\triangle ABC$, D and E are points on the sides AB and AC respectively such that $DE \parallel BC$. (i) If $\frac{AD}{DB} = \frac{3}{4}$ and $AC = 15$ cm find AE . **JUL-21**

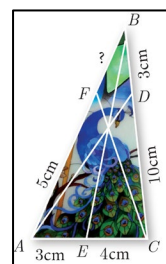
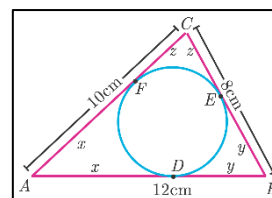


10. Check whether AD is bisector of $\angle A$ of $\triangle ABC$ in each of the following
 - (i) $AB = 5$ cm, $AC = 10$ cm, $BD = 1.5$ cm and $CD = 3.5$ cm. **JUL-20**
 - (ii) $AB = 4$ cm, $AC = 6$ cm, $BD = 1.6$ cm and $CD = 2.4$ cm. **JUL-23**
11. In $\triangle ABC$, D and E are points on the sides AB and AC respectively. Show that $DE \parallel BC$ if $AB = 12$ cm, $AD = 8$ cm, $AE = 12$ cm and $AC = 18$ cm.
12. What length of ladder is needed to reach a height of 7 ft along the wall when the base of the ladder is 4 ft from the wall? Round off your answer to the next tenth place. **MDL-19**
13. A man goes 18 m due east and then 24 m due north. Find the distance of his current position from the starting point? **APR-24, JUL-22**
14. Find the length of the tangent drawn from a point whose distance from the centre of a circle is 5 cm and radius of the circle is 3 cm.
15. In Fig.4.65, $\triangle ABC$ is circumscribing a circle. Find the length of BC . **JUL-25**
16. If radii of two concentric circles are 4 cm and 5 cm then find the length of the chord of one circle which is a tangent to the other circle.
17. The length of the tangent to a circle from a point P , which is 25 cm away from the centre is 24 cm. What is the radius of the circle?



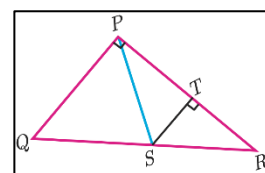
10 th MATHS PUBLIC EXAM PREPARATION-2026

18. A circle is inscribed in $\triangle ABC$ having sides 8 cm, 10 cm and 12 cm as shown in figure, Find AD , BE and CF .
19. A tangent ST to a circle touches it at B . AB is a chord such that $\angle ABT = 65^\circ$. Find $\angle AOB$, where " O " is the centre of the circle.
20. In two concentric circles, a chord of length 16 cm of larger circle becomes a tangent to the smaller circle whose radius is 6 cm. Find the radius of the larger circle.
21. An artist has created a triangular stained glass window and has one strip of small length left before completing the window. She needs to figure out the length of left out portion based on the lengths of the other sides as shown in the figure.
22. State Ceva's Theorem.**(Also Cevian)
23. State Menelaus's Theorem.**

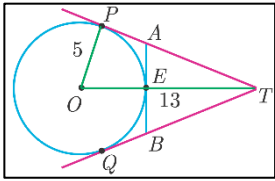


5 MARK IMPORTANT QUESTIONS

1. STATE AND PROVE THALES THEOREM OR BPT. APR-22, 24
2. STATE AND PROVE ANGLE BISECTOR THEOREM OR ABT. APR-23, 25, MDL-19, JUL-22
3. STATE AND PROVE PYTHAGORAS THEOREM OR BAUDHAYANA THEOREM. JUL-23, 24, 25, SEP-21
4. STATE AND PROVE ALTERNATE SEGMENT OR CHORD TANGENT THEOREM. PTA-4
5. Two poles of height ' a ' metres and ' b ' metres are ' p ' metres apart. Prove that the height of the point of intersection of the lines joining the top of each pole to the foot of the opposite pole is given by $\frac{ab}{a+b}$ metres. APR-23
6. Two triangles QPR and QSR , right angled at P and S respectively are drawn on the same base QR and on the same side of QR . If PR and SQ intersect at T , prove that $PT \times TR = ST \times TQ$. PTA-6
7. $ABCD$ is a trapezium in which $AB \parallel DC$ and P, Q are points on AD and BC respectively, such that $PQ \parallel DC$ if $PD = 18$ cm, $BQ = 35$ cm and $QC = 15$ cm, find AD . JUL-22
8. In figure $\angle QPR = 90^\circ$, PS is its bisector. If $ST \perp PR$, PTA-2
Prove that $ST \times (PQ + PR) = PQ \times PR$.
9. $ABCD$ is a quadrilateral in which $AB = AD$, the bisector of $\angle BAC$ and $\angle CAD$ intersect the sides BC and CD at the points E and F respectively. Prove that $EF \parallel BD$.
10. P and Q are the mid-points of the sides CA and CB respectively of a $\triangle ABC$, right angled at C . Prove that $4(AQ^2 + BP^2) = 5AB^2$. APR-22
11. An Aeroplane after take off from an airport and flies due north at a speed of 1000 km/hr. At the same time, another aeroplane take off from the same airport and flies due west at a speed of 1200 km/hr. How far apart will be the two planes after $1/2$ hours? APR-22
12. The hypotenuse of a right triangle is 6 m more than twice of the shortest side. If the third side is 2 m less than the hypotenuse, find the sides of the triangle. PTA-3
13. 5 m long ladder is placed leaning towards a vertical wall such that it reaches the wall at a point 4 m high. If the foot of the ladder is moved 1.6 m towards the wall, then find the distance by which the top of the ladder would slide upwards on the wall.



10 th MATHS PUBLIC EXAM PREPARATION-2026

14. The perpendicular PS on the base QR of a $\triangle PQR$ intersects QR at S , such that $QS = 3SR$. Prove that $2PQ^2 = 2PR^2 + QR^2$ **PTA-6**
15. PQ is a chord of length 8 cm to a circle of radius 5 cm. The tangents at P and Q intersect at a point T . Find the length of the tangent TP . **MDL-19**
16. In figure, O is the centre of the circle with radius 5 cm. T is a point such that $OT = 13$ cm and OT intersects the circle E , if AB is the tangent to the circle at E , find the length of AB .
- 
17. **Show that the angle bisectors of a triangle are concurrent.** **PTA-4**
18. A man whose eye-level is 2 m above the ground wishes to find the height of a tree. He places a mirror horizontally on the ground 20 m from the tree and finds that if he stands at a point C which is 4 m from the mirror B , he can see the reflection of the top of the tree. How high is the tree?
19. Two trains leave a railway station at the same time. The first train travels due west and the second train due north. The first train travels at a speed of 20 km/hr and the second train travels at 30 km/hr. After 2 hours, what is the distance between them?
20. An Emu which is 8 feet tall is standing at the foot of a pillar which is 30 feet high. It walks away from the pillar. The shadow of the Emu falls beyond Emu. What is the relation between the length of the shadow and the distance from the Emu to the pillar?

CHAPTER - 5. COORDINATE GEOMETRY

PUBLIC EXAM IMPORTANT QUESTIONS-2026

2 MARK IMPORTANT QUESTIONS

1. Find the area of the triangle whose vertices are $(-3,5)$, $(5,6)$ and $(5,-2)$
2. Show that the points $P(-1.5,3)$, $Q(6,-2)$, $R(-3,4)$ are collinear. **APR-22, PTA-4**
3. If the area of the triangle formed by the vertices $A(-1,2)$, $B(k,-2)$ and $C(7,4)$ (taken in order) is 22 sq. units, find the value of k . **JUL-22**
4. If the points $P(-1,-4)$, $Q(b,c)$ and $R(5,-1)$ are collinear and if $2b + c = 4$, then find the values of b and c . **JUL-21**
5. Find the area of the triangle formed by the points
(i) $(1,-1)$, $(-4,6)$ and $(-3,-5)$ (ii) $(-10,-4)$, $(-8,-1)$ and $(-3,-5)$
6. Determine whether the sets of points are collinear? (i) $(-\frac{1}{2}, 3)$, $(-5,6)$ and $(-8,8)$
7. Find the value of 'a' for which the given points are collinear. (i) $(2,3)$, $(4,a)$ and $(6,-3)$
8. If the points $A(-3,9)$, $B(a,b)$ and $C(4,-5)$ are collinear and if $a + b = 1$, then find a and b . **APR-24**

10 th MATHS PUBLIC EXAM PREPARATION-2026

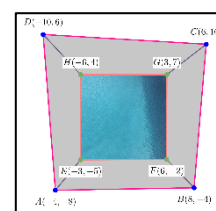
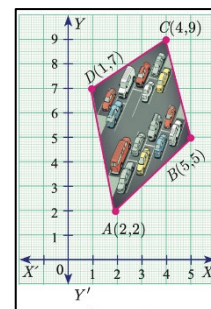
9. What is the slope of a line whose inclination is 30° ?
10. What is the inclination of a line whose slope is $\sqrt{3}$?
11. Find the slope of a line joining the given points
(i) $(-6,1)$ and $(-3,2)$ **JUL-25** (ii) $(-\frac{1}{3}, \frac{1}{2})$ and $(\frac{2}{7}, \frac{3}{7})$ (iii) $(14,10)$ and $(14, -6)$ **JUL-20**
12. The line r passes through the points $(-2,2)$ and $(5,8)$ and the line s passes through the points $(-8,7)$ and $(-2,0)$. Is the line r perpendicular to s ?
13. The line p passes through the points $(3, -2)$, $(12,4)$ and the line q passes through the points $(6, -2)$ and $(12,2)$. Is p parallel to q ? **JUL-22,24, APR-22**
14. Show that the points $(-2,5)$, $(6, -1)$ and $(2,2)$ are collinear.
15. What is the inclination of a line whose slope is 1 **PTA-3**
16. Find the slope of a line joining the points
(i) $(5, \sqrt{5})$ with the origin **JUL-22, 23** (ii) $(\sin \theta, -\cos \theta)$ and $(-\sin \theta, \cos \theta)$ **PTA-2**
17. Show that the given points are collinear: $(-3, -4)$, $(7,2)$ and $(12,5)$ **JUL-21**
18. If the three points $(3, -1)$, $(a, 3)$ and $(1, -3)$ are collinear, find the value of a .
19. The line through the points $(-2, a)$ and $(9,3)$ has slope $-\frac{1}{2}$. Find the value of a . **APR-25**
20. The line through the points $(-2,6)$ and $(4,8)$ is perpendicular to the line through the points $(8,12)$ and $(x, 24)$. Find the value of x . **PTA-6**
21. Find the equation of a straight line whose
(i) Slope is 5 and y intercept is -9 (ii) Inclination is 45° and y intercept is 11
22. Calculate the slope and y intercept of the straight line $8x - 7y + 6 = 0$ **JUL-21**
23. Find the equation of a line passing through the point $(3, -4)$ and having slope $\frac{-5}{7}$.
24. Find the equation of a line passing through the point $A(1,4)$ and perpendicular to the line joining points $(2,5)$ and $(4,7)$.
25. Find the equation of a straight line passing through $(5, -3)$ and $(7, -4)$.
26. Find the equation of a line which passes through $(5,7)$ and makes intercepts on the axes equal in magnitude but opposite in sign.
27. Find the intercepts made by the line $4x - 9y + 36 = 0$ on the coordinate axes.
28. The equation of a straight line is $2(x - y) + 5 = 0$. Find its slope, inclination and intercept on the Y axis.
29. Find the equation of a line whose inclination is 30° and making an intercept -3 on the Y axis.
30. Find the slope and y intercept of $\sqrt{3}x + (1 - \sqrt{3})y = 3$.
31. Find the value of ' a ', if the line through $(-2,3)$ and $(8,5)$ is perpendicular to $y = ax + 2$

10 th MATHS PUBLIC EXAM PREPARATION-2026

32. The hill in the form of a right triangle has its foot at (19,3). The inclination of the hill to the ground is 45° . Find the equation of the hill joining the foot and top. **PTA-2**
33. A cat is located at the point $(-6, -4)$ in xy plane. A bottle of milk is kept at (5,11). The cat wish to consume the milk travelling through shortest possible distance. Find the equation of the path it needs to take its milk. **APR-23**
34. Find the equation of a straight line which has slope $\frac{-5}{4}$ and passing through the point $(-1,2)$. **APR-22,24**
35. Find the equation of a line through the given pair of points
(i) $(2, \frac{2}{3})$ and $(\frac{-1}{2}, -2)$ (ii) $(2,3)$ and $(-7, -1)$
36. Find the intercepts made by the following lines on the coordinate axes.
(i) $3x - 2y - 6 = 0$ **JUL-21** (ii) $4x + 3y + 12 = 0$ **JUL-25**
37. Find the slope of the straight line $6x + 8y + 7 = 0$. **JUL-24**
38. Show that the straight lines $2x + 3y - 8 = 0$ and $4x + 6y + 18 = 0$ are parallel.
39. Show that the straight lines $x - 2y + 3 = 0$ and $6x + 3y + 8 = 0$ are perpendicular. **APR-25, PTA-5**
40. Find the equation of a straight line which is parallel to the line $3x - 7y = 12$ and passing through the point (6,4). **JUL-23**
41. Find the slope of the line which is
(i) parallel to $y = 0.7x - 11$ (ii) perpendicular to the line $x = -11$
42. Check whether the given lines are parallel or perpendicular
(i) $\frac{x}{3} + \frac{y}{4} + \frac{1}{7} = 0$ and $\frac{2x}{3} + \frac{y}{2} + \frac{1}{10} = 0$
(ii) $5x + 23y + 14 = 0$ and $23x - 5y + 9 = 0$
43. If the straight lines $12y = -(p + 3)x + 12$, $12x - 7y = 16$ are perpendicular then find ' p '. **APR-23**
44. Find the equations of the lines, whose sum and product of intercepts are 1 and -6 respectively
45. Find the image of the point (3,8) with respect to the line $x + 3y = 7$ assuming the line to be a plane mirror.
-

5 MARK IMPORTANT QUESTIONS

- Find the area of the quadrilateral formed by the points $(8,6)$, $(5,11)$, $(-5,12)$ and $(-4,3)$. **JUL-22, APR-23**
- The given diagram shows a plan for constructing a new parking lot at a campus. It is estimated that such construction would cost ₹ 1300 per square feet. What will be the total cost for making the parking lot?
- Find the area of the quadrilateral whose vertices are at **APR-24, JUL-23,24, 25**
 - $(-9, -2)$, $(-8, -4)$, $(2,2)$ and $(1, -3)$
 - $(-9,0)$, $(-8,6)$, $(-1, -2)$ and $(-6, -3)$
- Find the value of k , if the area of a quadrilateral is 28 sq.units, whose vertices are taken in the order $(-4, -2)$, $(-3, k)$, $(3, -2)$ and $(2,3)$ **APR-25, JUL-20, PTA-5**
- In the figure, the quadrilateral swimming pool shown is surrounded by concrete patio. Find the area of the patio. **PTA-2**
- Without using Pythagoras theorem, show that the points $(1, -4)$, $(2, -3)$ and $(4, -7)$ form a right angled triangle. **PTA-4**
- Show that the given points form a right angled triangle
 - $A(1, -4)$, $B(2, -3)$ and $C(4, -7)$
 - $L(0,5)$, $M(9,12)$ and $N(3,14)$
- Show that the given points form a parallelogram $A(2.5,3.5)$, $B(10, -4)$, $C(2.5, -2.5)$ and $D(-5,5)$.
- If the points $A(2,2)$, $B(-2, -3)$, $C(1, -3)$ and $D(x, y)$ form a parallelogram then find the value of x and y .
- Let $A(3, -4)$, $B(9, -4)$, $C(5, -7)$ and $D(7, -7)$. Show that $ABCD$ is a trapezium.
- A quadrilateral has vertices at $A(-4, -2)$, $B(5, -1)$, $C(6,5)$ and $D(-7,6)$. Show that the mid-points of its sides form a parallelogram. **APR-22**
- A line makes positive intercepts on coordinate axes whose sum is 7 and it passes through $(-3,8)$. Find its equation. **APR-25**
- If the vertices of a $\triangle ABC$ are $A(6,2)$, $B(-5, -1)$ and $C(1,9)$ through the vertex A **JUL-21, PTA-6**
 - find the equation of median
 - find the equation of altitude
- Find the equation of a straight line
 - passing through $(1, -4)$ and has intercepts which are in the ratio 2: 5 **MDL-19**
 - passing through $(-8,4)$ and making equal intercepts on the coordinate axes
- Find the equation of a straight line parallel to Y axis and passing through the point of intersection of the lines $4x + 5y = 13$ and $x - 8y + 9 = 0$.
- Find the equation of a line passing through $(6, -2)$ and perpendicular to the line joining the points $(6,7)$ and $(2, -3)$.
- $A(-3,0)$, $B(10, -2)$ and $C(12,3)$ are the vertices of $\triangle ABC$. Find the equation of the altitude through A and B . **JUL-23, APR-24**
- Find the equation of the perpendicular bisector of the line joining the points $A(-4,2)$ and $B(6, -4)$.
- Find the equation of a straight line through the intersection of lines $7x + 3y = 10$, $5x - 4y = 1$ and parallel to the line $13x + 5y + 12 = 0$ **JUL-24**
- Find the equation of a straight line joining the point of intersection of $3x + y + 2 = 0$ and $x - 2y - 4 = 0$ to the point of intersection of $7x - 3y = -12$ and $2y = x + 3$ **PTA-3**



10 th MATHS PUBLIC EXAM PREPARATION-2026

21. The area of a triangle is 5 sq.units. Two of its vertices are (2,1) and (3, -2). The third vertex is (x, y) where $y = x + 3$. Find the coordinates of the third vertex. **PTA-1**
 22. If vertices of a quadrilateral are at $A(-5,7)$, $B(-4,k)$, $C(-1, -6)$ and $D(4,5)$ and its area is 72 sq. units. Find the value of k .
 23. Without using distance formula, show that the points $(-2, -1)$, $(4,0)$, $(3,3)$ and $(-3,2)$ are vertices of a parallelogram.
 24. Find the equation of a line passing through the point of intersection of the lines $4x + 7y - 3 = 0$ and $2x - 3y + 1 = 0$ that has equal intercepts on the axes.
-

CHAPTER - 6. TRIGONOMETRY

PUBLIC EXAM IMPORTANT QUESTIONS-2026

2 MARK IMPORTANT QUESTIONS

1. Prove that $\tan^2 \theta - \sin^2 \theta = \tan^2 \theta \sin^2 \theta$ **JUL-23**
2. Prove that $\sec \theta - \cos \theta = \tan \theta \sin \theta$
3. Prove that $\sqrt{\frac{1+\cos \theta}{1-\cos \theta}} = \operatorname{cosec} \theta + \cot \theta$ **APR-24, MDL-19**
4. Prove that $\frac{\sec \theta}{\sin \theta} - \frac{\sin \theta}{\cos \theta} = \cot \theta$ **APR-23**
5. Prove that $1 + \frac{\cot^2 \theta}{1+\operatorname{cosec} \theta} = \operatorname{cosec} \theta$
6. Prove the following identities.
(i) $\cot \theta + \tan \theta = \sec \theta \operatorname{cosec} \theta$ (ii) $\tan^4 \theta + \tan^2 \theta = \sec^4 \theta - \sec^2 \theta$
7. Prove the following identities.
(i) $\frac{1-\tan^2 \theta}{\cot^2 \theta - 1} = \tan^2 \theta$ **JUL-24** (ii) $\frac{\cos \theta}{1+\sin \theta} = \sec \theta - \tan \theta$ **JUL-25**
8. Prove the following identities.
(i) $\sqrt{\frac{1+\sin \theta}{1-\sin \theta}} = \sec \theta + \tan \theta$ **JUL-20** (ii) $\sqrt{\frac{1+\sin \theta}{1-\sin \theta}} + \sqrt{\frac{1-\sin \theta}{1+\sin \theta}} = 2 \sec \theta$ **JUL-23**
9. A tower stands vertically on the ground. From a point on the ground, which is 48 m away from the foot of the tower, the angle of elevation of the top of the tower is 30° . Find the height of the tower. **PTA-1**
10. A kite is flying at a height of 75 m above the ground. The string attached to the kite is temporarily tied to a point on the ground. The inclination of the string with the ground is 60° . Find the length of the string, assuming that there is no slack in the string. **JUL-21,22**
11. Find the angle of elevation of the top of a tower from a point on the ground, which is 30 m away from the foot of a tower of height $10\sqrt{3}$ m. **JUL-21, 22, PTA-2**
12. A road is flanked on either side by continuous rows of houses of height $4\sqrt{3}$ m with no space in between them. A pedestrian is standing on the median of the road facing a row house. The angle of elevation from the pedestrian to the top of the house is 30° . Find the width of the road.

10 th MATHS PUBLIC EXAM PREPARATION-2026

13. A player sitting on the top of a tower of height 20 m observes the angle of depression of a ball lying on the ground as 60° . Find the distance between the foot of the tower and the ball. ($\sqrt{3} = 1.732$) **PTA-3**
14. From the top of a rock $50\sqrt{3}$ m high, the angle of depression of a car on the ground is observed to be 30° . Find the distance of the car from the rock. **PTA-6, APR-22,25**

5 MARK IMPORTANT QUESTIONS

1. If $\cos \theta + \sin \theta = \sqrt{2} \cos \theta$, then prove that $\cos \theta - \sin \theta = \sqrt{2} \sin \theta$
2. If $\operatorname{cosec} \theta + \cot \theta = P$, then prove that $\cos \theta = \frac{P^2 - 1}{P^2 + 1}$
3. Prove that $\left(\frac{\cos^3 A - \sin^3 A}{\cos A - \sin A} \right) - \left(\frac{\cos^3 A + \sin^3 A}{\cos A + \sin A} \right) = 2 \sin A \cos A$. **PTA-6**
4. If $\frac{\cos^2 \theta}{\sin \theta} = p$ and $\frac{\sin^2 \theta}{\cos \theta} = q$, then prove that $p^2 q^2 (p^2 + q^2 + 3) = 1$
5. Prove the following identities. $\frac{\sin^3 A + \cos^3 A}{\sin A + \cos A} + \frac{\sin^3 A - \cos^3 A}{\sin A - \cos A} = 2$
6. If $\sqrt{3} \sin \theta - \cos \theta = 0$, then show that $\tan 3\theta = \frac{3 \tan \theta - \tan^3 \theta}{1 - 3 \tan^2 \theta}$ **PTA-3**
7. If $\frac{\cos \theta}{1 + \sin \theta} = \frac{1}{a}$, then prove that $\frac{a^2 - 1}{a^2 + 1} = \sin \theta$
8. If $\cot \theta + \tan \theta = x$ and $\sec \theta - \cos \theta = y$, then prove that $(x^2 y)^{\frac{2}{3}} - (xy^2)^{\frac{2}{3}} = 1$.
9. If $\sin \theta + \cos \theta = p$ and $\sec \theta + \operatorname{cosec} \theta = q$, then prove that $q(p^2 - 1) = 2p$.
10. Two ships are sailing in the sea on either sides of a lighthouse. The angle of elevation of the top of the lighthouse as observed from the ships are 30° and 45° respectively. If the lighthouse is 200 m high, find the distance between the two ships. ($\sqrt{3} = 1.732$) **PTA-5, APR-24**
11. From a point on the ground, the angles of elevation of the bottom and top of a tower fixed at the top of a 30 m high building are 45° and 60° respectively. Find the height of the tower. ($\sqrt{3} = 1.732$) **APR-22**
12. To a man standing outside his house, the angles of elevation of the top and bottom of a window are 60° and 45° respectively. If the height of the man is 180 cm and if he is 5 m away from the wall, what is the height of the window? ($\sqrt{3} = 1.732$) **JUL-22**
13. A statue 1.6 m tall stands on the top of a pedestal. From a point on the ground, the angle of elevation of the top of the statue is 60° and from the same point the angle of elevation of the top of the pedestal is 40° . Find the height of the pedestal. ($\tan 40^\circ = 0.8391$, $\sqrt{3} = 1.732$)
14. From the top of a tower 50 m high, the angles of depression of the top and bottom of a tree are observed to be 30° and 45° respectively. Find the height of the tree. ($\sqrt{3} = 1.732$)
15. As observed from the top of a 60 m high lighthouse from the sea level, the angles of depression of two ships are 28° and 45° . If one ship is exactly behind the other on the same side of the lighthouse, find the distance between the two ships. ($\tan 28^\circ = 0.5317$) **PTA-1**
16. From the top of the tower 60 m high the angles of depression of the top and bottom of a vertical lamp post are observed to be 38° and 60° respectively. Find the height of the lamp post. ($\tan 38^\circ = 0.7813$, $\sqrt{3} = 1.732$) **JUL-20, MDL-19**
17. An aeroplane at an altitude of 1800 m finds that two boats are sailing towards it in the same direction. The angles of depression of the boats as observed from the aeroplane are 60° and 30° respectively. Find the distance between the two boats. ($\sqrt{3} = 1.732$)

10 th MATHS PUBLIC EXAM PREPARATION-2026

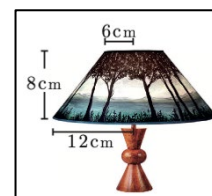
18. From the top of a lighthouse, the angle of depression of two ships on the opposite sides of it are observed to be 30° and 60° . If the height of the lighthouse is h meters and the line joining the ships passes through the foot of the lighthouse, show that the distance between the ships is $\frac{4h}{\sqrt{3}}$ m. **APR-23**
19. From the top of a 12 m high building, the angle of elevation of the top of a cable tower is 60° and the angle of depression of its foot is 30° . Determine the height of the tower. **JUL-25**
20. A pole 5 m high is fixed on the top of a tower. The angle of elevation of the top of the pole observed from a point 'A' on the ground is 60° and the angle of depression to the point 'A' from the top of the tower is 45° . Find the height of the tower. ($\sqrt{3} = 1.732$) **JUL-24**
21. From the top of a tree of height 13 m the angle of elevation and depression of the top and bottom of another tree are 45° and 30° respectively. Find the height of the second tree. ($\sqrt{3} = 1.732$)
22. The angles of elevation and depression of the top and bottom of a lamp post from the top of a 66 m high apartment are 60° and 30° respectively. Find
(i) The height of the lamp post. (ii) The difference between height of the lamp post and the apartment.
(iii) The distance between the lamp post and the apartment.
($\sqrt{3} = 1.732$)
23. Two ships are sailing in the sea on either side of the lighthouse. The angles of depression of two ships as observed from the top of the lighthouse are 60° and 45° respectively. If the distance between the ships is $200\left(\frac{\sqrt{3}+1}{\sqrt{3}}\right)$ metres, find the height of the lighthouse.
24. A building and a statue are in opposite side of a street from each other 35 m apart. From a point on the roof of building the angle of elevation of the top of statue is 24° and the angle of depression of base of the statue is 34° . Find the height of the statue. ($\tan 24^\circ = 0.4452$, $\tan 34^\circ = 0.6745$). **PTA-4**

CHAPTER - 7. MENSURATION**PUBLIC EXAM IMPORTANT QUESTIONS--2026****2 MARK IMPORTANT QUESTIONS**

1. A cylindrical drum has a height of 20 cm and base radius of 14 cm . Find its curved surface area and the total surface area. **JUL-22, 24**
2. The curved surface area of a right circular cylinder of height 14 cm is 88 cm^2 . Find the diameter of the cylinder. **JUL-23, 25**
3. A garden roller whose length is 3 m long and whose diameter is 2.8 m is rolled to level a garden. How much area will it cover in 8 revolutions?
4. The radius of a conical tent is 7 m and the height is 24 m . Calculate the length of the canvas used to make the tent if the width of the rectangular canvas is 4 m ? **APR-23, PTA-3**
5. If the total surface area of a cone of radius 7 cm is 704 cm^2 , then find its slant height. **JUL-22**
6. Find the diameter of a sphere whose surface area is 154 m^2 . **JUL-20**
7. The radius of a spherical balloon increases from 12 cm to 16 cm as air being pumped into it. Find the ratio of the surface area of the balloons in the two cases. **APR-22**
8. If the base area of a hemispherical solid is 1386 sq. metres, then find its total surface area? **APR-24**
9. The slant height of a frustum of a cone is 5 cm and the radii of its ends are 4 cm and 1 cm . Find its curved surface area. **APR-25, JUL-25**
10. A sphere, a cylinder and a cone are of the same height which is equal to its radius, where as cone and cylinder are of same height. Find the ratio of their curved surface areas.
11. The ratio of the radii of two right circular cones of same height is 1: 3. Find the ratio of their curved surface area when the height of each cone is 3 times the radius of the smaller cone. **PTA-2**
12. The radius and height of a cylinder are in the ratio 5: 7 and its curved surface area is 5500sq. cm . Find its radius and height. **APR-23, JUL-22**
13. A solid iron cylinder has total surface area of 1848 sq.cm . Its curved surface area is five - sixth of its total surface area. Find the radius and height of the iron cylinder.
14. 4 persons live in a conical tent whose slant height is 19 m . If each person require 22 m^2 of the floor area, then find the height of the tent.
15. Find the volume of a cylinder whose height is 2 m and whose base area is 250 m^2 . **APR-24, 25, JUL-21**
16. The volume of a solid right circular cone is 11088 cm^3 . If its height is 24 cm then find the radius of the cone. **JUL-23, PTA-2**
17. The ratio of the volumes of two cones is 2: 3. Find the ratio of their radii if the height of second cone is double the height of the first.
18. The volumes of two cones of same base radius are 3600 cm^3 and 5040 cm^3 . Find the ratio of heights. **JUL-24, APR-22, PTA-4**
19. If the ratio of radii of two spheres is 4: 7, find the ratio of their volumes. **APR-23, 25**
20. A cone of height 24 cm is made up of modeling clay. A child reshapes it in the form of a cylinder of same radius as cone. Find the height of the cylinder. **PTA-3**
21. Find the Volume of the iron used to make a hollow cylinder of height 9 cm and whose internal and external radii are 21 cm and 28 cm respectively.

5 MARK IMPORTANT QUESTIONS

1. An industrial metallic bucket is in the shape of the frustum of a right circular cone whose top and bottom diameters are 10 m and 4 m and whose height is 4 m . Find the curved and total surface area of the bucket.
2. A girl wishes to prepare birthday caps in the form of right circular cones for her birthday party, using a sheet of paper whose area is 5720 cm^2 , how many caps can be made with radius 5 cm and height 12 cm .
3. The radius of a sphere increases by 25%. Find the percentage increase in its surface area.
MDL-19
4. The internal and external diameters of a hollow hemispherical vessel are 20 cm and 28 cm respectively. Find the cost to paint the vessel all over at ₹ 0.14 per cm^2 .
JUL-24
5. The frustum shaped outer portion of the table lamp has to be painted including the top part. Find the total cost of painting the lamp if the cost of painting 1sq. cm is ₹ 2 .
6. Calculate the mass of a hollow brass sphere if the inner diameter is 14 cm and thickness is 1 mm , and whose density is 17.3 g/cm^3 . (Hint: mass = density $\times$ volume) **MDL-19**
7. If the radii of the circular ends of a frustum which is 45 cm high are 28 cm and 7 cm , find the volume of the frustum. **APR-24, JUL-21, PTA-5**
8. A cylindrical glass with diameter 20 cm has water to a height of 9 cm . A small cylindrical metal of radius 5 cm and height 4 cm is immersed completely. Calculate the raise of the water in the glass?
JUL-20
9. If the circumference of a conical wooden piece is 484 cm then find its volume when its height is 105 cm . **JUL-22**
10. A solid sphere and a solid hemisphere have equal total surface area. Prove that the ratio of their volume is $3\sqrt{3}:4$. **PTA-6**
11. A container open at the top is in the form of a frustum of a cone of height 16 cm with radii of its lower and upper ends are 8 cm and 20 cm respectively. Find the cost of milk which can completely fill a container at the rate of ₹ 40 per litre. **APR-22**
12. A toy is in the shape of a cylinder surmounted by a hemisphere. The height of the toy is 25 cm Find the total surface area of the toy if its common diameter is 12 cm .
JUL-21, 25
13. Arul has to make arrangements for the accommodation of 150 persons for his family function. For this purpose, he plans to build a tent which is in the shape of cylinder surmounted by a cone. Each person occupies 4 sq. m of the space on ground and 40 cu. meter of air to breathe. What should be the height of the conical part of the tent if the height of cylindrical part is 8 m ? **APR-23**
14. A funnel consists of a frustum of a cone attached to a cylindrical portion 12 cm long attached at the bottom. If the total height be 20 cm , diameter of the cylindrical portion be 12 cm and the diameter of the top of the funnel be 24 cm . Find the outer surface area of the funnel.
PTA-1
15. Nathan, an engineering student was asked to make a model shaped like a cylinder with two cones attached at its two ends. The diameter of the model is 3 cm and its length is 12 cm . If each cone has a height of 2 cm , find the volume of the model that Nathan made. **APR-22, 25**



10 th MATHS PUBLIC EXAM PREPARATION-2026

16. A vessel is in the form of a hemispherical bowl mounted by a hollow cylinder. The diameter is 14 cm and the height of the vessel is 13 cm . Find the capacity of the vessel.
17. From a solid cylinder whose height is 2.4 cm and the diameter 1.4 cm , a cone of the same height and same diameter is carved out. Find the volume of the remaining solid to the nearest cm^3 .
18. A capsule is in the shape of a cylinder with two hemisphere stuck to each of its ends. If the length of the entire capsule is 12 mm and the diameter of the capsule is 3 mm , how much medicine it can hold?
19. A metallic sphere of radius 16 cm is melted and recast into small spheres each of radius 2 cm . How many small spheres can be obtained? **APR-25, JUL-23, 24**
20. A right circular cylindrical container of base radius 6 cm and height 15 cm is full of ice cream. The ice cream is to be filled in cones of height 9 cm and base radius 3 cm , having a hemispherical cap. Find the number of cones needed to empty the container. **APR-24, PTA-5, 6**
21. An aluminium sphere of radius 12 cm is melted to make a cylinder of radius 8 cm . Find the height of the cylinder. **JUL-25**
22. A conical flask is full of water. The flask has base radius r units and height h units, the water is poured into a cylindrical flask of base radius xr units. Find the height of water in the cylindrical flask.
23. Water is flowing at the rate of 15 km per hour through a pipe of diameter 14 cm into a rectangular tank which is 50 m long and 44 m wide. Find the time in which the level of water in the tanks will rise by 21 cm .
24. A solid sphere of radius 6 cm is melted into a hollow cylinder of uniform thickness. If the external radius of the base of the cylinder is 5 cm and its height is 32 cm , then find the thickness of the cylinder.
25. Find the number of coins, 1.5 cm in diameter and 2 mm thick, to be melted to form a right circular cylinder of height 10 cm and diameter 4.5 cm . **PTA-1**
26. A metallic sheet in the form of a sector of a circle of radius 21 cm has central angle of 216° . The sector is made into a cone by bringing the bounding radii together. Find the volume of the cone formed. **PTA-2**
27. A hollow metallic cylinder whose external radius is 4.3 cm and internal radius is 1.1 cm and whole length is 4 cm is melted and recast into a solid cylinder of 12 cm long. Find the diameter of solid cylinder.

CHAPTER - 8. STATISTICS & PROBABILITY

PUBLIC EXAM IMPORTANT QUESTIONS -2026

2 MARK IMPORTANT QUESTIONS

- Find the range and coefficient of range of the following data: 25,67,48,53,18,39,44. **APR-24**
- Find the range of the following distribution. **APR-25, PTA-5**

Age (in years)	16 – 18	18 – 20	20 – 22	22 – 24	24 – 26	26 – 28
Number of students	0	4	6	8	2	2

- The range of a set of data is 13.67 and the largest value is 70.08 . Find the smallest value. **PTA-4**
- Find the range and coefficient of range of the following data.
(i) 63,89,98,125,79,108,117,68 **APR-23, JUL-20** (ii) 43.5,13.6,18.9,38.4,61.4,29.8
- If the range and the smallest value of a set of data are 36.8 and 13.4 respectively, then find the largest value. **JUL-2**
- Calculate the range of the following data.

Income	400 – 450	450 – 500	500 – 550	550 – 600	600 – 650
Number of workers	8	12	30	21	6

- Find the standard deviation of first 21 natural numbers. **JUL-23, 24, PTA-6**
- The mean of a data is 25.6 and its coefficient of variation is 18.75. Find the standard deviation. **PTA-3**
- The standard deviation and mean of a data are 6.5 and 12.5 respectively. Find the coefficient of variation. **MDL-19**
- If the mean and coefficient of variation of a data are 15 and 48 respectively, then find the value of standard deviation.
- If $n = 5$, $\bar{x} = 6$, $\sum x^2 = 765$, then calculate the coefficient of variation.
- Two coins are tossed together. What is the probability of getting different faces on the coins? **APR-22, JUL-25**
- What is the probability that a leap year selected at random will contain **53 Saturdays**.
- What is the probability that a **non-leap** year selected at random will contain **53 Saturdays**.
- A die is rolled and a coin is tossed simultaneously. Find the probability that the die shows an odd number and the coin shows a head. **JUL-21,22**
- Express the sample space for rolling two dice using tree diagram.
- Write the sample space for selecting two balls at a time from a bag containing 6 balls numbered 1 to 6 (using tree diagram). **PTA-4**
- A coin is tossed thrice. What is the probability of getting two consecutive tails? **JUL-24**
- If $P(A) = 0.37$, $P(B) = 0.42$, $P(A \cap B) = 0.09$ then find $P(A \cup B)$.
- A flower is selected at random from a basket containing 80 yellow, 70 red and 50 white flowers. Find the probability of selecting a yellow or red flower? **APR-25**
- Two dice are rolled together. Find the probability of getting a doublet or sum of faces as 4.

10 th MATHS PUBLIC EXAM PREPARATION-2026

22. A and B are two candidates seeking admission to IIT. The probability that A getting selected is 0.5 and the probability that both A and B getting selected is 0.3 . Prove that the probability of B being selected is atmost 0.8 . **APR-23, PTA-6**
23. If $P(A) = \frac{2}{3}, P(B) = \frac{2}{5}, P(A \cup B) = \frac{1}{3}$ then find $P(A \cap B)$. **PTA-1.**
24. If A and B are two mutually exclusive events of a random experiment and $P(\text{not } A) = 0.45, P(A \cup B) = 0.65$, then find $P(B)$. **PTA-2**
25. The probability that atleast one of A and B occur is 0.6 . If A and B occur simultaneously with probability 0.2 , then find $P(\bar{A}) + P(\bar{B})$.
26. If the range and coefficient of range of the data are 20 and 0.2 respectively, then find the largest and smallest values of the data.
27. If two dice are rolled, then find the probability of getting the product of face value 6 or the difference of face values 5 .
28. In a two children family, find the probability that there is at least one girl in a family.
29. A bag contains 5 white and some black balls. If the probability of drawing a black ball from the bag is twice the probability of drawing a white ball then find the number of black balls.
30. In a box there are 20 non-defective and some defect bulbs. If the probability that a bulb selected at random from the box found to be defective is $\frac{3}{8}$ then, find the number of defective bulbs.

5 MARK IMPORTANT QUESTIONS-2026

- The number of televisions sold in each day of a week are 13, 8, 4, 9, 7, 12, 10 . Find its standard deviation.
- The marks scored by 10 students in a class test are 25, 29, 30, 33, 35, 37, 38, 40, 44, 48. Find the standard deviation.
- Find the mean and variance of the first n natural numbers.
- The marks scored by the students in a slip test are given below. Find the standard deviation of their marks. **PTA-1, JUL-25**

x	4	6	8	10	12
f	7	3	5	9	5

- Marks of the students in a particular subject of a class are given below. Find its standard deviation.

Marks	0 – 10	10 – 20	20 – 30	30 – 40	40 – 50	50 – 60	60 – 70
Number of students	8	12	17	14	9	7	4

- A teacher asked the students to complete 60 pages of a record note book. Eight students have completed only 32,35,37,30,33,36,35 and 37 pages. Find the standard deviation of the pages completed by them. **JUL-24**
- The rainfall recorded in various places of five districts in a week are given below. Find its standard deviation.

10 th MATHS PUBLIC EXAM PREPARATION-2026

Rainfall (in mm)	45	50	55	60	65	70
Number of places	5	13	4	9	5	4

8. The time taken by 50 students to complete a 100 meter race are given below. Find its standard deviation. **PTA-5**

Time taken (sec)	8.5 – 9.5	9.5 – 10.5	10.5 – 11.5	11.5 – 12.5	12.5 – 13.5
Number of students	6	8	17	10	9

9. The following table gives the values of mean and variance of heights and weights of the 10th standard students of a school. **PTA-6**

	Height	Weight
Mean	155 cm	46.50 kg
Variance	72.25 cm ²	28.09 kg

Which is more varying than the other?

- Find the coefficient of variation of 24,26,33,37,29,31. **APR-24, JUL-23, MDL-19**
- The time taken (in minutes) to complete a homework by 8 students in a day are given by 38,40,47,44,46,43,49,53. Find the coefficient of variation.
- The total marks scored by two students Sathya and Vidhya in 5 subjects are 460 and 480 with standard deviation 4.6 and 2.4 respectively. Who is more consistent in performance? **APR-25**
- Two dice are rolled. Find the probability that the sum of outcomes is **JUL-21, 24, 25**
 - equal to 4
 - greater than 10
 - less than 13
- A die is rolled and a coin is tossed simultaneously. Find the probability that the die shows an odd number and the coin shows a head. **JUL-21,22**
- A bag contains 6 green balls, some black and red balls. Number of black balls is as twice as the number of red balls. Probability of getting a green ball is thrice the probability of getting a red ball. Find (i) number of black balls (ii) total number of balls.
- If A is an event of a random experiment such that $P(A):P(\bar{A}) = 17:15$ and $n(S) = 640$ then find
 - $P(\bar{A})$
 - $n(A)$. **PTA-3**
- Two unbiased dice are rolled once. Find the probability of getting **APR-23, JUL-20,22**
 - a doublet (equal numbers on both dice)
 - the product as a prime number
 - the sum as a prime number
 - the sum as 1
- Three fair coins are tossed together. Find the probability of getting **PTA-5**
 - all heads
 - atleast one tail
 - atmost one head
 - atmost two tails
- A bag contains 5 red balls, 6 white balls, 7 green balls, 8 black balls. One ball is drawn at random from the bag. Find the probability that the ball drawn is **APR-25, JUL-22**
 - white
 - black or red
 - not white
 - neither white nor black
- Two dice are rolled together. Find the probability of getting a doublet or sum of faces as 4.
- In a class of 50 students, 28 opted for NCC, 30 opted for NSS and 18 opted both NCC and NSS. One of the students is selected at random. Find the probability that **APR-22, PTA-1, 4**

10 th MATHS PUBLIC EXAM PREPARATION-2026

- (i) The student opted for NCC but not NSS.
(ii) The student opted for NSS but not NCC.
(iii) The student opted for exactly one of them.
23. In an apartment, in selecting a house from door numbers 1 to 100 randomly, find the probability of getting the door number of the house to be an even number or a perfect square number or a perfect cube number.
24. Two dice are rolled once. Find the probability of getting an even number on the first die or a total of face sum 8. **APR-24,JUL-23**
25. A coin is tossed thrice. Find the probability of getting exactly two heads or atleast one tail or two consecutive heads. **PTA-2, 6**
26. Three unbiased coins are tossed once. Find the probability of getting atleast 2 tails or atleast 2 heads.
27. A box contains cards numbered 3,5,7,9, ... 35,37. A card is drawn at random from the box. Find the probability that the drawn card have either multiples of 7 or a prime number.
28. In a class of 35 , students are numbered from 1 to 35 . The ratio of boys to girls is 4: 3. The roll numbers of students begin with boys and end with girls. Find the probability that a student selected is either a boy with prime roll number or a girl with composite roll number or an even roll number.
29. Prices of peanut packets in various places of two cities are given below. In which city, prices were more stable?

Prices in city A	20	22	19	23	16
Prices in city B	10	20	18	12	15

PUBLIC EXAM IMPORTANT QUESTIONS-2026

8 MARK IMPORTANT QUESTIONS (GEOMETRY & GRAPH)

1. Construct a triangle similar to a given triangle **PQR** with its sides equal to $\frac{3}{5}$ of the corresponding sides of the triangle **PQR** (Scale Factor $\frac{3}{5} < 1$). JUL-23
2. Construct a triangle similar to a given triangle **LMN** with its sides equal to $\frac{4}{5}$ of the corresponding sides of the triangle **LMN** (Scale Factor $\frac{4}{5} < 1$).
3. Construct a triangle similar to a given triangle **PQR** with its sides equal to $\frac{7}{4}$ of the corresponding sides of the triangle **PQR** (Scale Factor $\frac{7}{4} > 1$). JUL-25
4. Construct a triangle similar to a given triangle **ABC** with its sides equal to $\frac{6}{5}$ of the corresponding sides of the triangle **ABC** (Scale Factor $\frac{6}{5} > 1$). JUL-20
5. Construct a triangle similar to a given triangle **PQR** with its sides equal to $\frac{7}{3}$ of the corresponding sides of the triangle **PQR** (Scale Factor $\frac{7}{3} > 1$). JUL-22
6. Construct a ΔPQR in which $PQ = 8 \text{ cm}$, $\angle R = 60^\circ$ and the Median **RG** from R to PQ is 5.8 cm. Find the length of the altitude from R to PQ. PTA-3
7. Construct a ΔPQR in which base $PQ = 4.5 \text{ cm}$, $\angle R = 35^\circ$ and the Median **RG** from R to PQ is 6 cm. APR-24, JUL-21
8. Construct a ΔPQR in which $QR = 5 \text{ cm}$, $\angle P = 40^\circ$ and the Median **PG** from P to QR is 4.4 cm. Find the length of the Altitude from P to QR. JUL-24
9. Construct a ΔPQR in which such that $QR = 5 \text{ cm}$, $\angle P = 30^\circ$ and the Altitude from P to QR is of length 4.2 cm PTA-5
10. Construct a ΔPQR in which such that $QR = 5.5 \text{ cm}$, $\angle P = 25^\circ$ and the Altitude from P to QR is of length 4 cm. APR-22
11. Draw a triangle ΔABC of base $BC = 8 \text{ cm}$, $\angle A = 60^\circ$ and the Bisector of $\angle A$ meets BC at D such that $BD = 6 \text{ cm}$. APR-23,25, MDL-19
12. Draw a triangle ΔABC of base $BC = 5.6 \text{ cm}$, $\angle A = 40^\circ$ and the Bisector of $\angle A$ meets BC at D such that $BD = 4 \text{ cm}$.
13. Draw a circle of diameter 6 cm from a point P, which is 8 cm away from its centre. Draw the two tangents **PA** and **PB** to the circle and measure their lengths. JUL-21, 22, PTA-6, APR-24
14. Draw the two tangents from a point which is 10 cm away from the centre of a circle of radius 5 cm. Also, measure the lengths of the tangents. JUL-20,23,24
15. Take a point which is 11 cm away from the centre of a circle of radius 4 cm and draw the two tangents to the circle from that point. APR-23, 25, PTA-2
16. Draw the two tangents from a point which is 5 cm away from the centre of a circle of diameter 6 cm. Also , measure the lengths of the tangents. APR-22, JUL-25

17. Draw a circle of **radius 4.5 cm**. Take a point on the circle. Draw the tangent at that point using the **Alternate Segment Theorem**.
18. Draw a circle of **radius 3 cm**. Take a point P on this circle and draw a **tangent at P**.

1. **Varshika drew 6 circles** with different sizes. Draw a graph for the relationship between the **diameter and circumference** (approximately related) of each circle as shown in the table and use it to find the circumference of a circle when its **diameter is 6 cm**. APR-23

Diameter (x) cm	1	2	3	4	5
Circumference (y) cm	3.1	6.2	9.3	12.4	15.5

2. A bus is travelling at a uniform speed of **50 km / hr**. Draw the **distance time graph** and hence find APR-25

- (i) The **constant of variation**.
- (ii) How far will it travel in **90 minutes or 1 ½ hrs?**
- (iii) The time required to cover a distance of **300 km** from the graph.

3. A garment shop announces a flat **50 % discount** on every purchase of items for their customers.

Draw the graph for the relation between the **Marked Price and the Discount**. Hence find

- (i) The marked price when a customer gets a **discount of ₹ 3250 (from graph)**.
- (ii) The discount when the **marked price is ₹ 2500**.

4. Graph the following **linear function** $y = \frac{1}{2}x$. Identify the constant of variation and verify it with the graph. Also find JUL-23

- (i) y when $x = 9$ and (ii) x when $y = 7.5$.

5. A two wheeler parking zone near bus stand charges as below.

Time x (in hours)	4	8	12	24
Amount y (in ₹)	60	120	180	360

Check if the amount charged are in direct variation or in inverse variation to the parking time.

Graph the data. Also

- (i) Find the amount to be paid when parking **time is 6 hrs**.
- (ii) Find the parking duration when the amount **paid is ₹ 150**.

6. **Nishanth** is the winner in a Marathon race **12 km distance**. He ran at the uniform speed of **12 km / hr** and reached the destination in **1 hour**. He was followed by Aradhana, Jeyanth, Sathya and Swetha with their respective speed of **6 km / hr, 4 km / hr, 3 km / hr and 2 km / hr**. And, they covered the distance in **2 hrs, 3 hrs, 4 hrs and 6 hours** respectively.

Draw the Speed- time graph and use it to find time taken to Kaushik with his speed **2.4 km / hr**.

10 th MATHS PUBLIC EXAM PREPARATION-2026

7. Draw the graph of $xy = 24, x, y > 0$. Using the graph find,

APR-24

- (i) y when $x = 3$ and (ii) x when $y = 6$.

8. The following table shows the data about the number of pipes and the time taken to fill the same tank.

JUL-24

No. of. pipes x	2	3	6	9
Time Taken y (in mins)	45	30	15	10

(i) Draw the graph for the above data and hence

(ii) Find the time taken to fill the tank when **five pipes** are used

9. Discuss the nature of solutions of the following quadratic equations.

(i) $x^2 + x - 12 = 0$ (ii) $x^2 - 8x + 16 = 0$

JUL-25

10. Graph the following quadratic equations and state the nature of solutions.

(i) $x^2 - 9x + 20 = 0$ (ii) $x^2 - 4x + 4 = 0$

APR-25, JUL-22

APR-22

(iii) $x^2 - 6x + 9 = 0$

11. Draw the graph of $y = 2x^2$ and hence solve $2x^2 - x - 6 = 0$.

PTA-4

12. Draw the graph of $y = x^2 + x - 2$ and hence solve $x^2 + x - 2 = 0$.

JUL-24, PTA-1

13. Draw the graph of $y = x^2 - 4x + 3$ and hence solve $x^2 - 6x + 9 = 0$.

JUL-22, APR-22

14. Draw the graph of $y = x^2 - 4$ and hence solve $x^2 - x - 12 = 0$.

15. Draw the graph of $y = x^2 + 3x + 2$ and hence solve $x^2 + 2x + 1 = 0$

PTA-5

16. Draw the graph of $y = x^2 + 3x - 4$ and hence solve $x^2 + 3x - 4 = 0$.

JUL-21

17. Draw the graph of $y = x^2 - 5x - 6$ and hence solve $x^2 - 5x - 14 = 0$.

APR-23, PTA-2, 6

18. Draw the graph of $y = 2x^2 - 3x - 5$ and hence solve $2x^2 - 4x - 6 = 0$.

APR-24, JUL-20, PTA-3